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POLYTECHNIQUE MONTRÉAL

affiliée à l'Université de Montréal

Analyzing Incident Effects on Subway Operation

MOSTAFA ABOLFAZLI

Département de mathématiques et de génie industriel

Mémoire présenté en vue de l'obtention du diplôme de *Maîtrise ès sciences appliquées*

Génie industriel

Mai 2024

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POLYTECHNIQUE MONTRÉAL

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Ce mémoire intitulé :

Analyzing Incident Effects on Subway Operation

présenté par **Mostafa ABOLFAZLI**

en vue de l'obtention du diplôme de *Maîtrise ès sciences appliquées*

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Julie CARREAU, membre

DEDICATION

To my beloved partner,

To my amazing parents,

And to myself….

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I am profoundly grateful to the individuals and organizations listed below, whose contributions have been indispensable in the completion of this research.

First and foremost, I extend my sincere appreciation to my supervisor, Professor Martin Trépanier, whose guidance and mentorship have been invaluable. His unwavering support, encouragement, and availability for insightful discussions have enriched my learning experience over the past two years.

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Last but not least, I express deep gratitude and appreciation to my beloved partner, whose support and encouragement have been my guiding light throughout this journey.

RÉSUMÉ

Les systèmes de métro jouent un rôle vital dans le transport urbain en fournissant des moyens de déplacement efficaces, fiables et accessibles pour des millions de personnes quotidiennement. Néanmoins, le grand nombre de passagers quotidiens et les intervalles de service fréquents exposent les systèmes de métro à des pannes de service et à des incidents. Les incidents de métro doivent être traités efficacement, car les retards causés par eux peuvent se propager rapidement à travers le réseau, perturbant plusieurs stations. Mesurer les impacts totaux des incidents sur l'exploitation est essentiel pour améliorer le temps de réponse et permettre des stratégies de récupération efficaces.

Les études existantes sur la gestion des incidents de métro se concentrent principalement sur le développement de modèles pour prédire la durée des incidents à l'aide des journaux d'incidents. Cependant, ces journaux sont sujets aux erreurs humaines et sont limités dans leur portée de surveillance. Par conséquent, il y a eu peu de recherches sur l'analyse des effets complets des incidents sur l'exploitation du métro. Pour combler cette lacune, cette recherche vise à proposer une méthode pour quantifier l'impact global des incidents en exploitant les enregistrements opérationnels des trains, puis en catégorisant les incidents à l'aide d'une méthode de regroupement. Le principal objectif de cette étude est d'identifier les modèles de gravité des incidents de métro afin que les exploitants de transport en commun puissent améliorer la manière dont ils mesurent et comprennent les effets complets des incidents.

Le cadre de recherche se compose de plusieurs sous-objectifs visant à caractériser les impacts des incidents dans le système de métro de Montréal. Cela implique l'utilisation de deux types de données : les données de positionnement des trains obtenues à partir du système de signalisation et les données de journaux d'incidents rapportées par l'exploitant. Dans le système de métro de Montréal, les informations de positionnement des trains sont dérivées d'un système de signalisation par bloc fixe couvrant toutes les voies du réseau. Ce système, en conjonction avec un mécanisme de pilotage automatique, facilite l'enregistrement précis des moments spécifiques où un train entre et sort d'un bloc, avec une précision jusqu'à la seconde. Par conséquent, l'objectif initial de cette étude est de développer un algorithme de fouille de données capable de gérer efficacement des données de blocs étendues et d'extraire des trajectoires de trains.

Le deuxième objectif de cette thèse est d'analyser les impacts des incidents sur l'exploitation du métro. Cela est réalisé en utilisant le temps d'occupation des stations, dérivé des données de trajectoires, comme mesure pour quantifier les retards des trains. Ensuite, les incidents et leurs retards associés sont identifiés dans les données de trajectoires en faisant référence aux informations des journaux d'incidents. Ensuite, des indicateurs de gravité, représentant l'étendue de l'impact de l'incident, sont formulés pour agréger les effets complets de chaque incident sur l'exploitation. Ces indicateurs comprennent le nombre de trains affectés, le nombre de paires train-station affectées, le temps de retard total et une variable binaire indiquant si les deux sens d'une ligne sont affectés par l'incident. Enfin, ces indicateurs de gravité sont utilisés comme entrée pour le regroupement en K-means afin de catégoriser les incidents en fonction de leurs niveaux d'impact.

Les principales conclusions de cette étude révèlent trois clusters distincts d'incidents avec des degrés variables d'impact sur l'exploitation du métro. Le cluster 1 est constitué d'incidents avec des impacts minimes, n'affectant qu'un seul sens d'une ligne de métro et entraînant un temps de retard total moyen de 16 minutes. Le cluster 2, qui représente la majorité des incidents, a eu des effets modérés sur les deux sens des lignes de métro, entraînant un temps de retard total moyen de 52 minutes et affectant environ 12 trains par incident. Le cluster 3 comprend des incidents avec des impacts graves, causant des perturbations significatives des services de métro en affectant en moyenne 26 trains et entraînant un temps de retard total moyen de 273 minutes. L'influence des heures de pointe sur la propagation des incidents était notable, les périodes de pointe du matin et du soir connaissant des nombres moyens plus élevés de trains affectés. Les heures de pointe, caractérisées par un intervalle de service réduit, ont entraîné des incidents affectant un plus grand nombre de trains par rapport aux heures creuses et aux week-ends. De plus, un examen des causes des incidents a révélé que les incidents les plus fréquents tendent à avoir des impacts modérés.

La comparaison entre les retards d'incidents signalés et ceux identifiés dans les données de trajectoire a mis en évidence un biais vers une surestimation des retards principaux dans les journaux d'incidents, en particulier pour les retards inférieurs à 20 minutes. Cela souligne l'importance de valider les données à partir de sources multiples pour améliorer l'exactitude des modèles prédictifs, en particulier pour les retards plus petits. De plus, l'existence de retards non étiquetés dépassant 5 minutes dans les données de trajectoire suggère des incidents potentiels manqués dans les journaux d'incidents, mettant en évidence les limites de se fier uniquement aux journaux d'incidents pour une analyse exhaustive des retards d'incidents.

Les résultats de cette recherche fournissent des informations précieuses sur la gestion des incidents de métro, posant les bases pour d'autres études visant à améliorer la performance des réseaux de transport en rail urbain lors de perturbations de service. La compréhension des modèles de gravité des incidents aide les exploitants de transport en commun à améliorer leur connaissance des impacts des incidents et à améliorer la prise de décision pour une allocation efficace des ressources et la gestion de l'exploitation du métro. En évaluant les modèles de gravité des incidents, les exploitants peuvent évaluer l'efficacité des stratégies de récupération et ajuster la réponse pour réduire les retards d'incidents.

ABSTRACT

Subway systems play a vital role in urban transportation by providing efficient, reliable, and accessible means of commuting for millions of people daily. Nevertheless, the large volume of daily passengers and the frequent service intervals expose subway systems to service failures and incidents. Subway incidents need to be effectively addressed, as delays caused by them can rapidly propagate across the network, disrupting multiple stations. Measuring the total impacts of incidents on operation is essential for improving response time and enabling efficient recovery strategies.

Existing studies on subway incident management are mainly focused on developing models to predict incident duration using incident logs. However, these logs are prone to human error and are constrained in their monitoring range. Consequently, there has been limited research on analyzing the full effects of incidents on subway operation. To bridge this gap, this research aims to propose a method for quantifying the overall impact of incidents by mining train operational records and then categorizing incidents using clustering method. The main goal of this study is to identify the severity patterns of subway incidents so as transit operators can enhance the way they measure and understand the full effects of incidents.

The research framework consists of several sub objectives aimed at characterizing the impacts of incidents within the Montreal subway system. This involves utilizing two types of data: train positioning data obtained from signaling system and incident logs data reported by the operator. In the Montreal subway system, train positioning information is derived from a fixed block signaling system covering all tracks in the network. This system, in conjunction with an autopilot mechanism, facilitates the precise logging of the specific times when a train enters and exits a block, with accuracy down to the second. Consequently, the initial goal of this study is to develop a data mining algorithm capable of efficiently handling extensive block data and extracting train trajectories.

The second objective of this thesis is to analyze the impacts of incidents on subway operation. This is accomplished by using station occupancy time, derived from trajectories data, as a measure to quantify train delays. Subsequently, incidents and their associated delays are identified within trajectories data by referencing information from incident logs. Following this, severity indicators ,representing the extent of incident impact, are formulated to aggregate the full effects of each incident on operation. These indicators include the number of affected trains, the number of affected train-station pairs, total delay time, and a binary variable indicating if both direction of a

line are affected by the incident. Finally, these severity indicators are utilized as input for K-means clustering to categorize incidents based on their impact levels.

The main findings reveal three distinct clusters of incidents with varying degrees of impact on subway operation. Cluster 1 consists of incidents with minimal impacts, affecting only one direction of a subway line and resulting in an average total delay time of 16 minutes. Cluster 2, which represents the majority of incidents, had moderate effects on both directions of subway lines, leading to an average total delay time of 52 minutes and affecting approximately 12 trains per incident. Cluster 3 includes incidents with severe impacts, causing significant disruptions to subway services by affecting an average of 26 trains and resulting in an average total delay time of 273 minutes. The influence of peak hours on incident propagation was noticeable, with morning and evening peak periods experiencing higher average numbers of affected trains. Peak hours, characterized by reduced service headway, resulted in incidents impacting a larger number of trains compared to off-peak hours and weekends. Additionally, an analysis of incident causes revealed that the majority of frequent incidents are categorized within cluster 2, tending to have moderate impacts.

The comparison between reported incident delays and those identified in trajectory data uncovered a potential bias towards overestimation of primary delays in incident logs, especially for delays less than 20 minutes. This highlights the importance of validating data from multiple sources to improve predictive models' accuracy, especially for smaller delays. Furthermore, the existence of unlabelled delays exceeding 5 minutes in trajectory data suggests potential incidents missed in incident logs, underscoring the limitations of solely relying on incident logs for a comprehensive incident delay analysis.

The results of this research provide valuable insights into subway incident management, laying the groundwork for further studies aimed at enhancing the performance of urban rail transit networks during service disruptions. Understanding incident severity patterns help transit operators to enhance their knowledge of incident impacts and improve decision-making for efficient resource allocation and subway operation management. By evaluating incident severity patterns, operators can assess the effectiveness of recovery strategies and adjust response to reduce incident delays.

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LIST OF SYMBOLS AND ABBREVIATIONS

AFT	Accelerated Time Failure
AP	Autopilot System
AVL	Automatic Vehicle Location
BD	Both Directions
CDV	Circuits de Voie
GPS	Global Positioning Systems
NAT	Number of Affected Trains
NATS	Number of Affected Train-Stations
PD	Primary Delay
RTSC	Railway and Transport Strategy Center
SD	Secondary Delay
SOT	Station Occupancy Time
STM	Société de Transport de Montréal
TDT	Total Delay Time

CHAPTER 1 INTRODUCTION

1.1 Background

Due to the population growth and increasing travel demands in large cities, subway systems are constantly expanding and have become a crucial mode of daily transportation. However, the large volume of daily commuters and the high frequency of services, impose significant challenges to the reliable operation of subway systems. These challenges emerge in the form of incidents that occasionally disrupt the service. A subway incident is an event that triggers service breakdown, resulting from various causes, including internal factors like system component failures (e.g., power outages) or external factors such as fires at stations or even a passenger falling ill on board (Weng et al., 2014)

The consequences of incidents on both passengers and system operators can be significant. Travel delays caused by incidents are the major consequence concerning passengers; they experience an increase in both waiting times at stations and travel time aboard the trains. Furthermore, train delays result in overcrowded platforms and carriages which reduce service quality (Bueno-Cadena et al., 2020; Weng et al., 2014). For the system operators, such unexpected events result in prolonged cycle times, requiring the need for increasing speeds and, consequently, energy consumption is risen to compensate for the lost time. Subsequently, modifications to the operating schedule and realignment of personnel and rolling stock utilization are necessary to accommodate these changes (Bueno-Cadena et al., 2020).

Subway incidents are critical and require an immediate response. Due to strong interaction and linkage effect between stations and lines in a subway network, even a small delay can rapidly propagate along the network, disrupting multiple stations and causing more delays, thereby affecting customer service negatively (Goverde, 2010). The delay of a subway train not only decreases the capacity of the line but also affects the succeeding trains entering the line. If the delay is transmitted to the terminal station, it affects the train turnaround, and the delay propagates to the other direction as well (Zhao & Ding, 2021). Thus, it is necessary to perform a comprehensive analysis of incident delays and its propagation range based on the characteristics of various incident types. In such a manner, operators will be provided with more information regarding the consequences of incidents, in order to minimize the overall service disturbances.

Previous studies on subway incident management are primarily focused on developing predictive models for incident duration using incident log data (Wang et al., 2023; Weng et al., 2015). Incident logs are manually recorded data which suffers from human errors and have limited monitoring range both in time and space. Although the existing models can fit incident data well, they don't provide any information about the network-wide effects of incidents such as the number of affected trains and total delay time. Since in most subway systems, the secondary delays of incidents are not explicitly recorded, it is impossible to infer how delays are developed and propagated along the network by using only incident data.

Recent advancement in communication systems and data collection technologies has offered new ways for the analysis of railway operation, providing deeper insight and more reliable information (Cerreto et al., 2018). The operational data collected from control and monitoring systems are a valuable resource to analyze actual train operations under service disruptions (Wen et al., 2019). Train operation records contain detailed information on timestamp and location of train movements along rail tracks. This data can be used to derive train trajectories and to analyze the propagation of incident delays across temporal and spatial dimensions. Therefore, analyzing delays and propagation patterns form historical records can provide more precise insights for railway operation managers.

1.2 Research Objective

The primary goal of this study is to propose an approach to identify severity patterns of subway incidents using clustering analysis of indicators calculated from operational records and incident log data obtained from the Montréal subway system. The research framework is designed around a set of specific steps which are referred to as sub-objectives, as follows:

1. Develop an algorithm to extract train trajectories from the operational records.
2. Define indicators related to delays at stations (severity indicators) to help identify abnormal operation events.
3. Propose a way to isolate incidents within trajectory data using incident logs data.
4. Develop and apply a clustering method that uses severity indicators to categorize incidents based on their impact.

1.3 Thesis Structure

This thesis consists of six chapters with this introduction being the first. Chapter 2 introduces the fundamental concepts of railway systems needed for understanding the remainder of the thesis. Additionally, a review of the most important and recent contributions in the literature related to approaches for mining operational data and incident management in railway systems is presented. This chapter concludes by highlighting gaps identified in the literature.

Chapter 3 describes the case study focusing on the Montréal subway network, detailing its underlying infrastructure. It further discusses the structure of the available datasets including train operational records and incident log reports.

In Chapter 4, the methodology is thoroughly explained covering preprocessing steps and input preparation for further analysis. The chapter outlines two stages of the methodological framework: extracting trajectories and measuring incident impacts from operational records are described in detail. The steps involved in each stage are explained in detail, and an example of the analyzed dataset for each step is provided to facilitate comprehension.

Chapter 5 presents the results of the research in detail. It starts with a descriptive analysis of the incident log data, followed by a descriptive analysis of the incident impact indicators. Subsequently, a comparison between the identified incident delays and reported incident delays is provided. Next, the chapter presents the findings of the clustering analysis on incident severity patterns, highlighting the unique characteristics of each cluster. Furthermore, it offers a temporal analysis of clusters and an investigation into incident causes.

Finally, chapter 6 summarizes the main findings and contributions of the thesis. Research limitations and directions for future research are also discussed.

CHAPTER 2 LITERATURE REVIEW

The research motivation and main objectives were described in the first chapter. Before presenting the main contributions of this research, it is important to define the issues resulting from subway incidents in more detail and review the existing contributions from the scientific community. In the first part of this chapter, the concepts of subway systems are presented. The basic definitions related to subway operation, signalling systems, incidents and disruptions, train delays, and traffic control are given. Then, the current practices of subway disruption management are presented, and the main problems are identified. A critical review of the existing approaches is provided, highlighting the gaps in the state-of-the-art models that are addressed by the method proposed in this thesis.

2.1 Basic Concepts of Subway Systems

In this section, the basic concepts of a subway system are introduced, and they will serve as the foundation for identifying the performance indicators that can be derived from train movement data.

A subway *line* describes a consecutive number of stations that are served by a train running between these stations. The first and last station of a subway line are called endpoints or *terminals*. Several lines together form a subway *network*. Different subway lines can be connected through *transfer* stations. The distance between trains in a metro system measured in time or space is called *headway* and *frequency* is the inverse of headway over a determined period. The *dwell time* of a train at a station is the time it spends without moving. *Peak hours* are the periods of day when there is the highest demand for transit services . These periods typically coincide with the morning and evening rush hours when people commute to and from work or school. Note that these definitions are motivated by the previous works in the literature(Guihaire & Hao, 2008; Kecman, 2014).

In this study, *station occupancy time* refers to the time difference between the arrival and departure of a train from a station. It includes the time to decelerate to a stop, door opening and closing time, dwell time and time to accelerate when leaving a station. In the following sections, the definitions related to the subway timetable, signalling system and subway traffic control are given.

2.1.1 Subway Timetable

A subway *timetable* is a schedule or plan that outlines the operating hours and frequency of metro train services on a specific route or network(Guihaire & Hao, 2008). It typically indicates the

arrival or/and departure time of the first and/or last trains at different stations along a route, as well as the frequency of service during different times of the day, such as peak hours or off-peak periods(Chai et al., 2022). The timetable helps passengers plan their journeys by providing information on when and where to catch a train, and when they can expect to arrive at their destination.

2.1.2 Signalling System

Safety and signalling systems are a crucial part of modern railways. They aim to guarantee the safety of train operations by averting derailments and collisions among trains utilizing the same infrastructure, as well as accidents involving trains and other vehicles or objects (Kecman, 2014). There are two types of signalling systems: fixed blocks and moving blocks. In a fixed block system, the line is divided into consecutive sections known as blocks. Each block section can be occupied by only one train at a time. The distance between two following trains is determined by adding the braking distance of the second train, the length of the block section, and an additional safety margin. Thus, the headway distance equals the headway of absolute braking distance plus the length of the block section. (Figure 2.1 (a))

On the other hand, moving block is based on absolute braking distance and the fixed block sections are eliminated. Absolute braking distance equals the braking distance of the second train plus an additional safety margin (Figure 2.1 (b)). If all trains travel at the same speed and have the same braking capacity, the separation distance between trains can, in theory, be only a few metres since it is impossible for the first train to stop faster than the one behind it. Therefore, it is not necessary to maintain a large safe distance between each two train. However, such a system requires a continuous detection of train movements.

In a fixed block system, blocks are used to give speed instructions, ensuring adequate space between trains to prevent potential overlaps. This signal is transmitted to the train operator, or directly to the train, by various means. It can be visual, through signs and lights, or communicated by radio to the operator by block stations. In modern systems, such as Montréal subway system, the signal is transmitted through track beacons and captured by beacon readers mounted under the train driver’s cab. The data transmitted to the train's electronic modules communicates speed details, and it's the responsibility of the train's automatic control system to adhere to the specified

speed. Figure 2.2 shows a train equipped with antennas receiving signals through beacons in a fixed block signalling system.

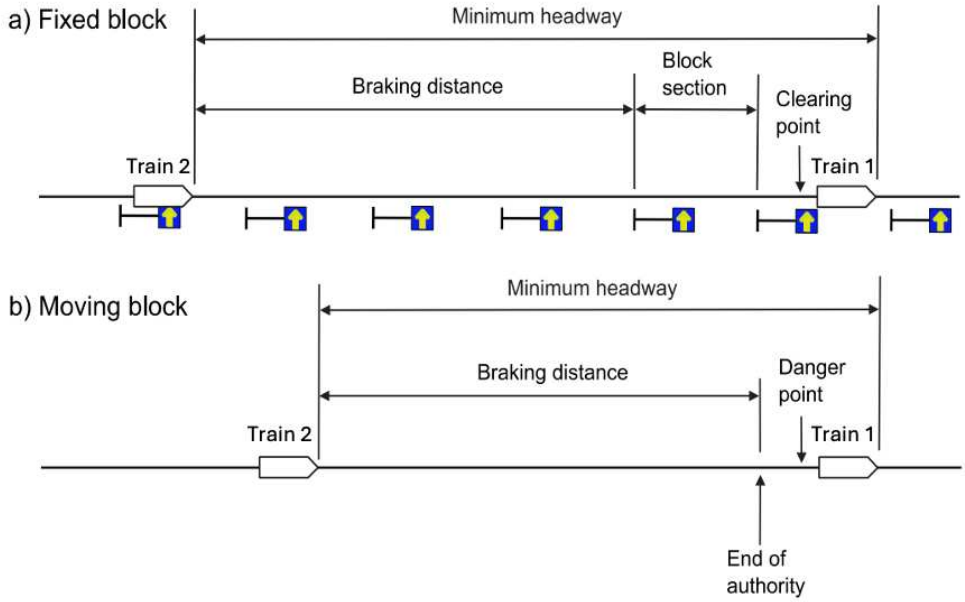


Figure 2.1 Principles of train separation in fixed block (a) and moving block (b) signalling system (Railway Signalling Principles, 2021)

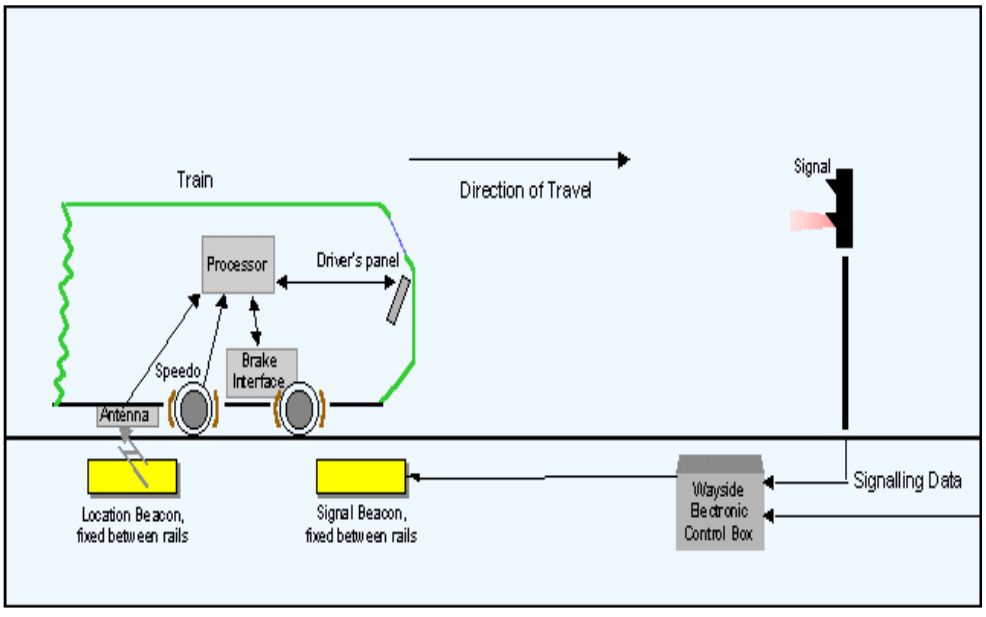


Figure 2.2 Schematic of a fixed block signalling system (Blais, 2007)

2.1.3 Track Clear Detection

The track-clear detection system is a basic element of signalling systems. The task of track-clear detection system is to check all relevant track sections are clear of vehicles before allowing a train to pass through. There are two types of track clear detection devices: track circuits and axle counters (Theeg & Vlasenko, 1997).

Track Circuit

A track circuit consists of an electrical circuit that includes the rails of a specific section. Typically, it has a current source at one end and a detection device at the other. Insulated rail joints divide the sections as shown in Figure 2.3. When a vehicle occupies the section, its axles create a short circuit by connecting the two rails. Consequently, the detection device does not receive any current, indicating that the section is occupied. This detection device is typically a track relay, which remains in a picked-up position when the section is clear and drops when the section is occupied. In modern setups, electronic detectors often replace relays. Since a track circuit operates on the closed-circuit principle, any interruption of the current ensures that the section is deemed occupied, leading to a safe state.

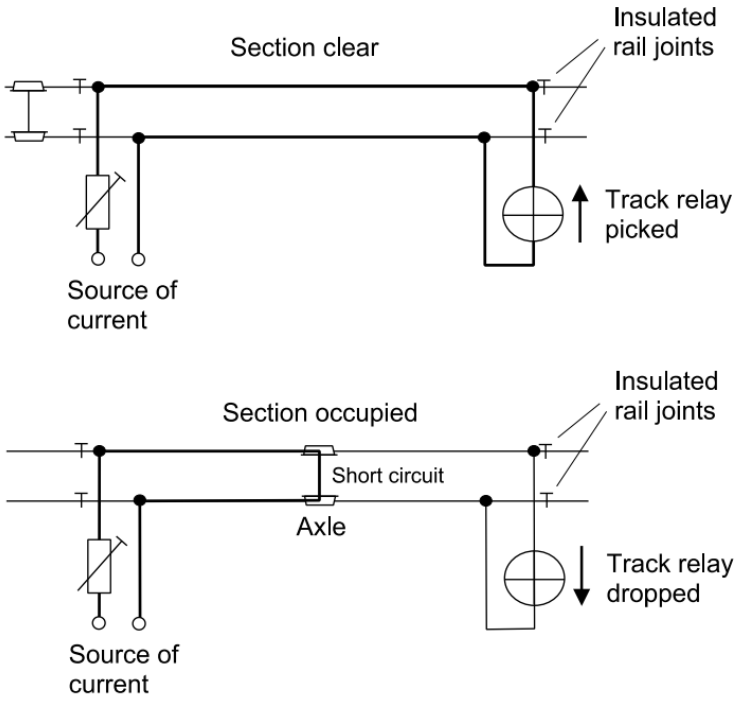


Figure 2.3 Track circuit (Railway Signalling Principles, 2021)

Axle counters

An axle counter comprises counting points positioned at the boundaries of a section, along with a connected counter (refer to Figure 2.4). A section occupancy is detected by comparing the number of axles entering the section with those leaving it. Clear indication requires parity of numbers. Counting points typically feature double contacts to discern the direction of movement, essential for accurate detection of axle entry or exit from the section.

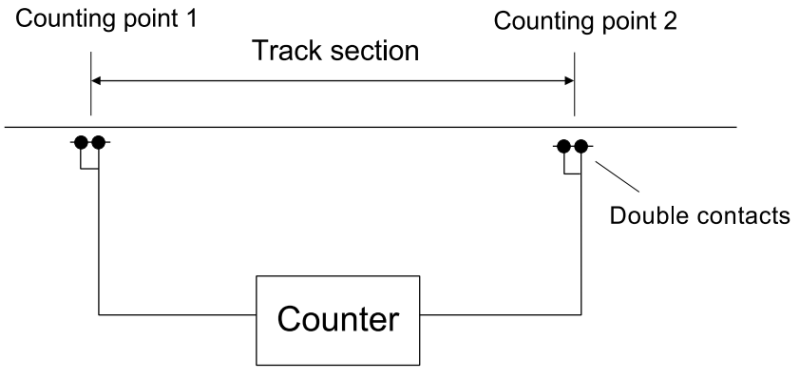


Figure 2.4 Axle counter (Railway Signalling Principles, 2021)

2.1.4 Blocking Time Theory

Blocking time theory is used to efficiently implement fixed block signalling in the railway traffic models (Hansen, 2008). As shown in Figure 2.5, blocking time refers to the duration that a specific block between two signals is exclusively allocated for the use of a single train, and therefore blocked for all other trains. It comprises the reaction time of the train driver, running time over the previous block, the running time of the current block, clearing time needed for the full train length to leave the block, and setup and release time of the signalling system.

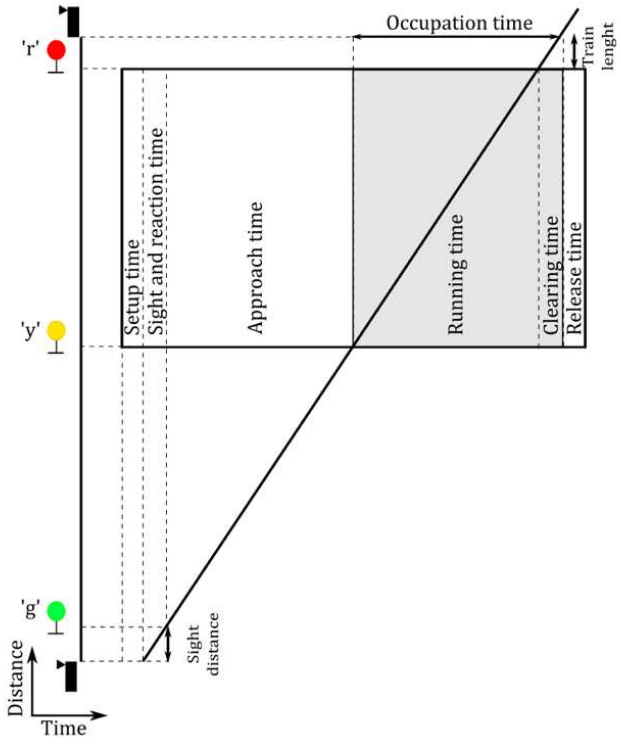


Figure 2.5 Schematic of blocking time in a space-time diagram (Kecman, 2014)

2.1.5 Train Positioning System

Keeping track of train positions is a necessity for the management of railway traffic. Traditionally, the monitoring of train positions was limited to only staffed stations and predetermined timetable points. However, with the emergence of advanced sensor and communication technologies, it is now possible to have a more detailed observation of moving trains (Kecman, 2014)

One of the most used systems for train positioning is train describers (Pachl, 2002). A train describer system keeps track of train positions over the route based on the messages received from the track-clear detection devices. These messages are continuously sent to the control centre to monitor train positions and are then archived for further investigation. This tracking acts as an automatic vehicle location (AVL) system which provides a significant amount of operational data at both temporal and spatial dimensions. By analyzing AVL data, it is possible to evaluate network performance according to different indicators in order to improve service reliability and thereby increase ridership. For the purpose of this study, the log archives of the Montréal subway train describer system have been made available by the Société de Transport de Montréal (STM).

2.1.6 Block Working Principles

An example of the block working principle is illustrated in Figure 2.6 . A train passes through signal 11 and enters the block section. The signal at the entrance of the section is locked in stop position to protect the train from following trains. This signal could be a lineside signal or simply a section limit (typically marked by block marker board), where a train must not proceed without cab signal indication. If multiple lines converge into the same section, the block locking applies to all signals leading into that block section. Once the train has completely exited the block section, and is protected by a stop signal, the block section is released. Subsequently, the signal at the entry point of the block section can be cleared for a following train.

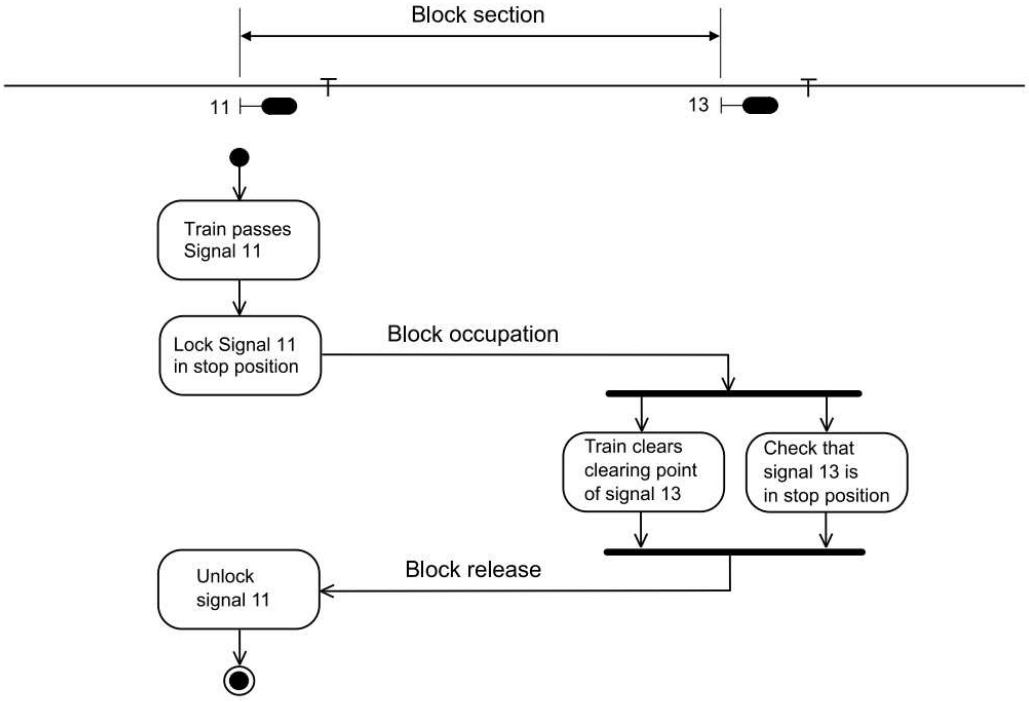


Figure 2.6 Block protection for train movements (Railway Signalling Principles, 2021)

2.1.7 Train Delays

Delays in railway systems occur because of the variability of process times (running time and dwell time), capacity and synchronization processes (Kecman, 2014). Small deviations from the scheduled (or planned) process times can be absorbed by the added time supplements and buffer times to the process times. However, major deviations of timetable caused by incidents need to be resolved by special disruption management strategies (Jespersen-Groth et al., 2009).

Delays resulting from incidents can be classified either as a primary delay (PD) or a secondary (knock-on) delay (Wen et al., 2017). PDs are independent form traffic control, and they are known as the source delays, while secondary delays are attributable to the propagation of PDs and are experienced by the following or even preceding trains. Extended running time of a train might lead to knock-on delays for subsequent trains. Likewise, increased dwell time at a station often leads to successive delays for other trains due to occupied platform tracks.

2.1.8 Disruption vs. Incident

Public transport systems can be affected by a wide range of events that might be unexpected, intentionally, or unintentionally caused by people or by a natural phenomenon. These incidents can range from adverse weather conditions to terrorist attacks, technical malfunctions, labour strikes, or the failure of physical components (Malandri et al., 2018). According to the Railway and Transport Strategy Center (RTSC), the most common incident causes can be classified into 6 groups which are shown in Table 2.1.

Although incidents often occur in railway systems, but not all of them disrupt the service. For instance, a shift replacement due to a driver's absence might be known as an incident, even if there were no service delays. A common approach for distinguishing disruptions from incidents is to define a delay threshold and use it as an identifier for disruption detection(Yap & Cats, 2019). Transit agencies around the world use different thresholds based on their level of service. In most subway systems, a disruption is defined as any kind of incidents resulting in a delay of 5 minutes or more (Barron et al., 2013). In several studies, the term disturbance was defined alongside with disruption (Cacchiani et al., 2014). Disturbances are defined as relatively small perturbations influencing the railway system, while disruptions are relatively large incidents leading to the closure of a railway line for a certain time. In (Cacchiani et al., 2014) authors declared that a

disturbance corresponds to the incidents in which certain railway processes including running time between two stations or dwelling at a station, last longer than what is planned by operators. In contrast, a disruption is a large incident which strongly influences the timetable, resulting in large delays of several trains. This might be caused by a temporary blockage of the railway, by infrastructure failures or by an accident.

Table 2.1 Common incident types

Incident Cause	**Example**
Rolling stock related	Problem with opening/closing doors
Signalling related	Interlocking failure
Equipment related	Power failure/ track failure
Staff related	Driver fault / shift absence
Passenger related	Sick passenger / suicide / violence
Other incidents	Police / security / fire

2.2 Mining Train Operational Data

The first step in developing a data-driven tool for monitoring train operations is to create a data mining procedure for processing train position records. The historical data sources utilized in current scientific literature range from manually recorded arrival and departure times at specific stations to comprehensive traffic realization data extracted from train describer log files (Kecman, 2014). Automatically collected railway operation data can be obtained from three distinct sources: infrastructure sensors, rolling stock sensors, and mobile GPS devices (Longo et al., 2012). A critical distinction between these various data collection methods lies in the type of data they gather. Traditional infrastructure-based sensors produce point-specific data, while rolling stock and mobile GPS sensors capture continuous data.

Infrastructure sensors, known as trackside devices, record train passages at specific timetabling points. These records are automatically gathered by train describer systems which keep track of trains' positions along their route (Longo et al., 2012). Train describer records are the most used data source for retrieving train trajectories on track section level (Kecman & Goverde, 2012a). The level of detail and the structure of the resulting data vary in different train describer systems. Consequently, a distinct approach is needed to design a data processing algorithm for each system. These processing tools are designed to refine the raw data and generate data structures suitable for performance analysis and the development of data-driven models.

Train describer data were archived primarily for maintenance and accident investigation purposes (Kecman, 2014). Their importance for performance analysis has been recognized in recent decades. Goverde and Hansen (2000) introduced a tool that links infrastructure messages with specific train numbers. Their algorithms associated infrastructure messages, which report changes in the status of signals, sections, and switches, with the corresponding train numbers responsible for these changes. By using this tool, the sequence of occupation and release time of blocks for each train run was obtained.

Later (Goverde & Meng, 2011) utilized the same information to identify delay chains and route conflicts. They constructed a system of delay trees to recognize the root causes of delays which allows the identification of individual delays into primary and secondary delays. Interestingly, they confirmed that dispatchers in the Netherlands inaccurately attributed over half of the delay causes. (Kecman & Goverde, 2012b) developed this model to include unmonitored line sections, where it is more challenging to differentiate between delays caused by signal-related issues and delays resulting from primary delays.

In another study (Labermeier, 2013), train describer data of the Swiss railway were processed for a comparative analysis of primary and secondary delays. It was indicated that the main cause of low punctuality levels in Switzerland is secondary delays resulting from waiting for delayed feeder trains. Sørensen et al. (2017) used operational data from the Norwegian railway to identify trains with primary delays and the trains affected by those delays. This analysis helps in identifying the source of delays, recognizing trains that often experience delays or cause them, and identifying stations where most of these delays occur.

2.3 Disruption Management in Railway Transport

The process of dealing with disruptions can be categorized along the timeline, with a particular focus on events that happen before and after a disruption. In this section, first this concept is explained in a general sense and then the practical examples in the context of public transportation are presented.

The hypothetical performance of a system under normal conditions and in the face of a disruption can be illustrated in Figure 2.7. The timeline on this graph highlights a risk event or disturbance occurring at the time "t_e". To meet certain service level agreements, reserve infrastructure and redundancy measures may be employed to maintain the system's performance level beyond the agreed-upon service levels. Before a disruption occurs, the system operates as planned. At the time a disruption occurs (time t_e), the performance of the system is degraded. The reaction and response to a disruption are aimed at implying recovery until the system is back to its normal state. Different terms such as vulnerability or survivability are used in this diagram which are briefly described in the following (Cats, 2016; Ge et al., 2022):

- *Reliability*: A public transport system is reliable if it can be trusted to work well or to behave in the way as intended.
- *Flexibility*: it relates to bending or pushing a system without breaking it. Other terms used for flexibility are adaptability, versatility, and adjustability.
- *Robustness:* it is the ability of a system to keep up to its functionality under the conditions that deviate from the normal state
- *Recovery*: is the process of bringing the system back to its original status after a disturbance or a disruption.
- *Resilience*: it is used to describe the ability of a system to withstands changes
- *Vulnerability*: is defined as the question to which extent extreme events can disrupt a transportation system.

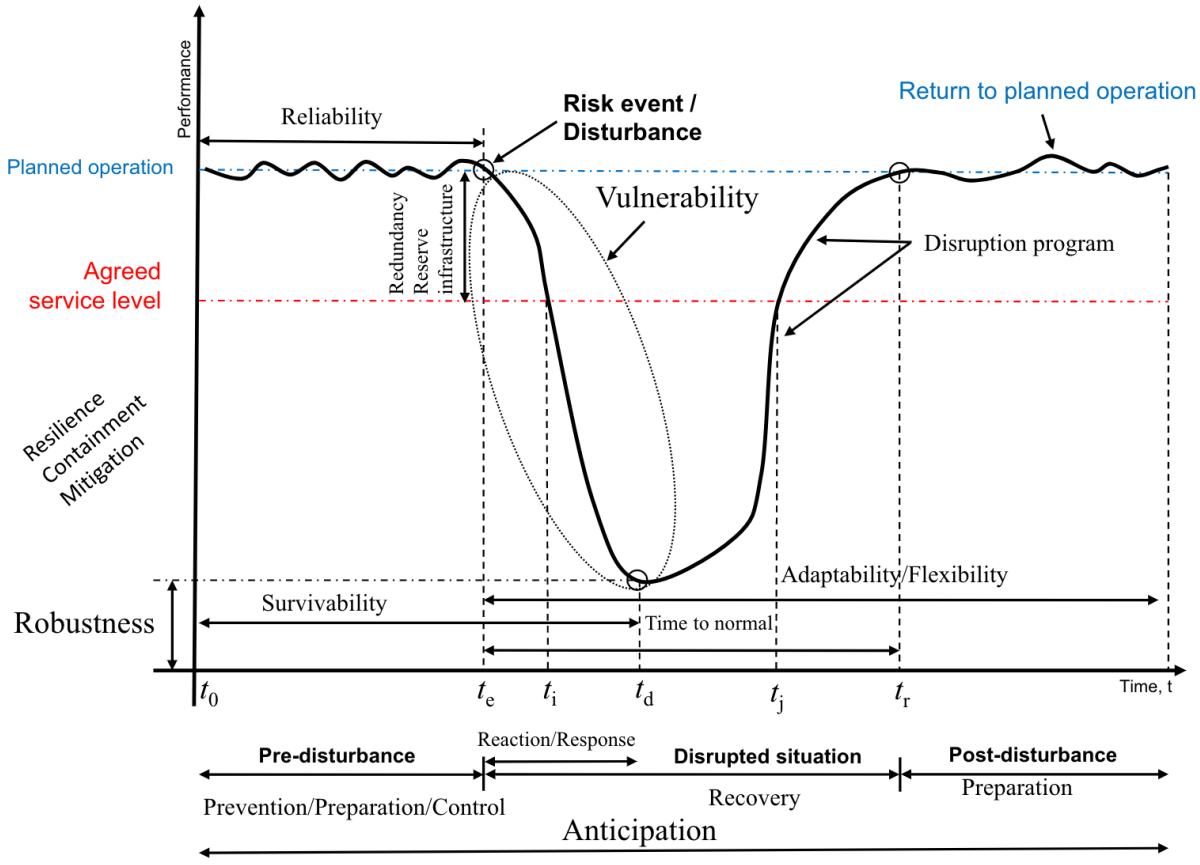


Figure 2.7 Performance diagram of a resilient system (Ge et al., 2022)

The strategies for handling public transport disruption can be classified into two main groups: proactive and reactive (Ge et al., 2022). Proactive approaches involve measures aimed at preventing and controlling disruptions, such as improved schedule synchronization and adding buffer times to encounter primary delays. In contrast, reactive approaches consist of responsive actions taken when disruptions occur, with the goal of minimizing their impact. These actions include emergency management, recovery, and delay management.

2.3.1 Delay Management

Disruptions can lead to delays which may cause passengers to arrive late at their intended destination or even earlier than scheduled at transfer stations. In literature, delay management refers to the challenge of determining whether connecting trains should wait for the delayed feeder train to be able to pick up late transfer passengers (König, 2020). If the connecting train is linked to other trains or transport modes, it might lead to a cascading or propagation problem that requires

appropriate management. That is why delay management is a part of disturbance management, not the other way around.

Different goals can be set by operators when dealing with delays; One goal is to return to the normal operation as quick as possible and avoid further delay propagation in the network. This is called train perspective with the objective of minimizing train delays (König, 2020). Another goal is to minimize delays for passengers or manage delays from a passenger perspective. It's worth mentioning that this study follows a train perspective goal due to lack of information regarding the number of daily passengers.

Current research on delay management can be categorized into three main aspects: analysis of delay causes, delay propagation analysis, and delay mitigation strategies (Wen et al., 2019). Previous studies shows that equipment, signal, and interlocking failures are the main causes of primary delay and operation interference is the main cause of secondary delays. Extensive research has been done on the analysis of different delay causes, such as delays due to extreme weather, fire, terrorist attack, suicide etc. (Weng et al., 2015, 2019). For instance, Chow et al. (2011) investigated the Honk Kong subway incidents, and explored causes related to fire and inadequate ventilation in order to raise public concerns on their consequences. In another study, authors categorized incidents from a high-speed railway in China into seven groups and established models for delay duration and delay influence due to each cause (Wen et al., 2020). The study revealed that the Log-normal probability distribution model provides a more accurate approximation of the observed distribution of delay durations for each cause.

Predicting and recovering from delays are the primary challenges in addressing the issue of delay propagation. Various factors contribute to delay propagation, including interaction among primary delays, knock-on delays, exogenous events, delay aggravation, and recovery. Probability functions can be employed to approximate these factors, considering elements from sections, stations, time, and train interactions (Wen et al., 2019).

Significant efforts have been dedicated to the estimation of train delays and their propagation within heavy railway systems. The proposed methods aim to estimate the expected primary or knock-on delays by utilizing various data sources such as predefined timetable, train movement data, railway infrastructure information and historical delay records. Yuan and Hansen (2007) introduced an analytical stochastic model aimed at estimating the propagation of train delays. The

primary challenge addressed in this model is how to estimate the convolution of individual independent distributions. Validation results demonstrate that the proposed analytical stochastic model accurately estimates the propagation of train delays, thereby improving the punctuality of train arrivals, departures, and knock-on delays. Kecman and Goverde (2015) proposed models developed by gathering running and dwell time data from a training set and establishing individual predictive models for estimating each type of process time. The findings of this model suggest that, irrespective of departure delays, the majority of running times show weak sensitivity to peak hours and no significant daily variation.

2.4 Subway Incident Management

Existing research on subway incident management primarily focuses on incident duration prediction using incident log data. For instance, Weng et al. (2014) developed an accelerated time failure (AFT) model with different parameters to explore the impacts of incidents on operation using incident records between 2005 and 2012 in Hong Kong. They showed that the log-logistic AFT model with random parameters achieved the best performance among all models. Louie et al. (2017) utilized incident records from Toronto's urban railway system to develop hazard-based models investigating the influences of causal and non-causal factors on disruption duration. Their findings indicated that disruption duration at transfer stations is shorter compared to non-transfer stations. Additionally, they observed no significant difference in disruption duration between peak and off-peak hours.

More recently, Wang et.al (2023) proposed a random survival forest model to predict and investigate factors influencing the duration of subway disruptions. Their findings indicated that their model outperformed the commonly used AFT models in terms of predictive accuracy. While the proposed models can fit incident data well, they don’t provide any information regarding the impact of each incident on operations, such as the number of affected trains and knock-on delays of subsequent trains. Incidents logs are reports of manual inspections and complaints from passengers which usually suffer from human errors and are restricted to a limited monitoring range in both time and space (Ji et al., 2018). Given the lack of details about knock-on delays in these reports, it becomes challenging to infer the mechanism of delay propagation along the subway line and network.

Another aspect of subway incident management studies involves analyzing the occurrence of specific incident causes. Mishara and Bardon (2017) conducted a comprehensive analysis of incidents associated with suicides in Canada. Their study offered valuable policy recommendations for preventing suicidal behavior, including the implementation of physical barriers. Yeo et al. (2006) claimed that subway incidents could also be attributed to the mental state of train drivers and mechanical issues. Diab and Shalaby (2020) employed statistical models to examine the effects of outdoor tracks and weather conditions on subway disruptions. Their findings revealed significant associations between these influencing factors and the frequency and duration of disruptions. Melo et al. (2011) analyzed the determinants of incident occurrence and highlighted that passenger demand and the level of service (e.g., headway during peak hours) significantly influence the frequency of incident occurrence.

Recent advancement in communication systems and data collection technologies has offered new ways for the analysis of subway operation, providing deeper insight and more reliable information (Cerreto et al., 2018). The operational data collected from control and monitoring systems are a valuable resource to analyze actual train operations under service disruptions (Wen et al., 2019). Train operation records contain detailed information on timestamp and location of train movements along rail tracks. This data can be used to derive train trajectories and to analyze the propagation of incident delays across temporal and spatial dimensions. The utilization of operational data for incident management studies in subway systems has been relatively rare in the literature. However, a recent study by Zhang et al. (2022) introduced a novel approach to identify service disruption and their propagation range, using trajectory data from a subway system. The authors assert that their proposed approach can detect 99% of all service disruptions including minor delays.

To conclude this section, a summary of the gaps identified in the existing literature is provided as follows:

- Current studies on subway incident analysis are solely based on incident logs data which are manually recorded reports and their validity is questionable.
- In metro systems, few studies have mined train trajectory data to detect primary and secondary delays resulting from incidents.
- Previous studies rarely pay attention to the overall impact of incidents along the network such as the number of affected trains and total delay time.

- Prior research aimed at classifying incidents based on their causes. However, this method appears to have a drawback, as certain causes sometimes have the same underlying mechanism on train operations.

It will be demonstrated in the following chapters that these gaps can be effectively addressed by applying data mining techniques to train operational data.

CHAPTER 3 CASE STUDY

In this chapter, a case study of the Montréal subway is introduced. The operation and infrastructure of the network are described in the first section. Then, the structure of the available datasets is presented.

3.1 The Montréal Metro

The Montréal metro is a rubber-tired underground rapid transit system serving the Greater Montréal area in Quebec, Canada. The metro is operated by the Société de transport de Montréal (STM), a public transport agency operating transit bus and rapid transit services in the urban agglomeration of Montréal. As of the third quarter of 2023, the Montréal metro ranks as the second-busiest rapid transit system in Canada in terms of daily ridership with serving an average of 938,100 unlinked passenger trips per weekday (APTA Q3, 2023).

The Montréal metro consists of four different lines which are identified by their colours, including orange, green, yellow, and blue, as shown in Figure 3.1. The four lines together have 68 stations spreading over 70 kilometres, serving the north, east and centre of the island of Montréal with connections to Longueuil with the yellow line, and Laval with the orange line. The orange line is the longest line with 31 stations and the yellow line is the shortest one with only 3 stations.

3.1.1 Infrastructure of the Montréal subway system

The Montréal Metro is the world's first subway system to operate entirely on rubber tires instead of steel wheels (Metrodemontreal.com). This feature not only provides a quieter ride for passengers and reduces noise for buildings above the tracks, but also facilitates easier navigation of slopes by trains. Unlike most subway systems, the Montréal metro is entirely underground. This design choice is influenced by Montréal's winter conditions, which would make rubber tire technology impractical for above-ground operation.



Figure 3.1 Schematic map of Montréal subway network (Société de transport de Montreal, n.d.)

Type of Trains

In this section, different types of trains used in the Montréal subway system are described (Société de transport de Montréal, n.d.):

- The MR-63 is the first generation of rubber-tired rolling stock of the Montréal metro. Based on the MP 59 of the Paris metro in France, the trains were in service from 1966 until 2018. In June 2018, the last of the MR-63 trains was completely retired after 53 years of service. An image of these metro cars is shown in Figure 3.2.
- The MR-73 is the second generation of rubber-tired rolling stock used in the Montréal metro system. Manufactured by Bombardier Transportation, they entered service in June 1976.

Since the retirement of the first generation MR-63s, the MR-73 trains are in use on all of Montréal's four Metro lines. An image of MR-73 is shown in Figure 3.3.

- The MPM-10, commonly known as the Azur, is the third and newest generation of rubber-tired rolling stock used in the Montréal metro, built by a consortium of Bombardier Transportation and Alstom. The first MPM-10 train entered service on the orange line in February 2016, replacing the entire first-generation MR-63 fleet. As of December 2021, 639 cars (71 sets) were in service on the orange and green Lines. Figure 3.4 shows an image of the MPM-10 trains. These trains are open gangway, allowing passengers to move between cars once on board such that the passenger load is more evenly distributed.



Figure 3.2 a MR-63 train at Beaugrand garage (wikipedia, 2023)



Figure 3.3 a MR-73 train arriving at Lucien-L'Allier station (Adqproductions, n.d.)



Figure 3.4 Two MPM-10 trains in Plamondon station (Dllu, n.d.)

Train Control System

The MR-73 and former MR-63 trains are equipped with a manual train control system, while the MPM-10 features automatic train control. In MR-73 trains, the operator manually opens and closes the doors and manages the traction/brake control system. Conversely, in MPM-10 trains, the operator can choose to operate the doors manually or allow them to function automatically. Upon pressing the departure button in the driver's cab ,the train initiates self-driving mode. The operator retains the option to manually operate the MPM-10 train as desired.

The Montreal subway operates on a fixed block signaling system, where rail tracks divided into blocks of variable length. The separation of each block is achieved by insulated rail joints. Speed instructions and station stop positions are transmitted by electrical induction through track circuits (circuits de voie (CDV)) embedded within the blocks. The configuration of signaling and block spacing adheres to the tunnel layout and profile, aiming to maintain a theoretical minimum interval of 90 seconds. Figure 3.5 shows an example of track circuits employed in the Montreal metro system.

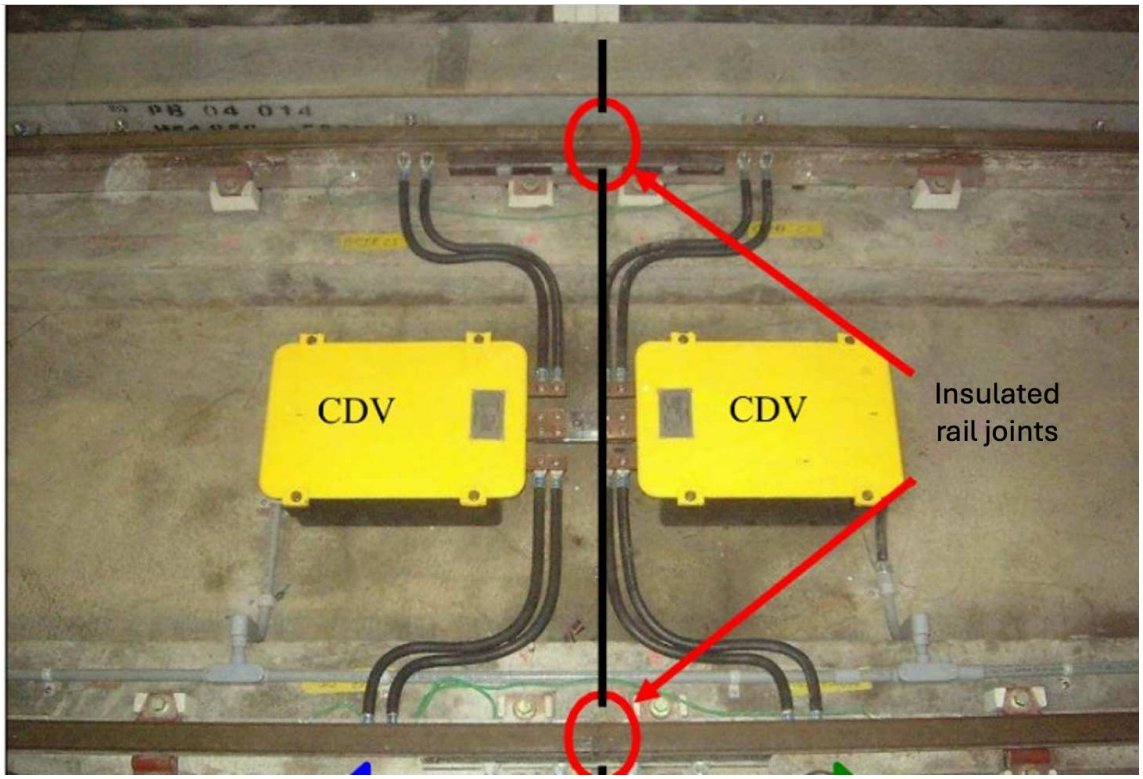


Figure 3.5 Track circuits in the Montréal metro (Blais, 2007)

3.1.2 Lines and Operation

Metro service starts at 5:30 AM, with the last trains departing between 00:30 and 01:00 AM on weekdays and Sunday, and between 01:00 to 01:30 AM on Saturday. The service headway ranges from 2 to 5 minutes for the peak hours (7 to 9 AM and 4 to 6 PM) to 4-10 minutes for the off-peak, and 6-12 on the weekends. Table 3.1 summarizes the operational statistics for different lines.

Table 3.1 Operational statistics for the Montréal subway lines

Line number	Colour	From	To	Length (km)	Number of Stations	Headway (min)		
						Peak	Off-peak	Weekend
1	Green	Angrignon	Honoré-Beaugrand	22.1	27	2-4	3-8	6-12
2	Orange	Côte-Vertu	Montmorency	30	31	2-7	2-8	6-10
4	Yellow	Berri–UQAM	Longueuil–Université-de-Sherbrooke	4.25	3	4-6	4-10	8-10
5	Blue	Snowdon	Saint-Michel	9.7	12	3-6	4-10	8-11

In the following section, we will describe the available datasets used for the purpose of this research.

3.2 Data

For this study, two different types of data are used including block occupation data and incident log data. The study period is from 1st January 2018 to 31st of October 2018. It should be noted that the main goal of this work is to understand the propagation pattern of incident delays along stations and a metro line. Thus, the yellow line is excluded from the analysis due to its limited coverage area.

3.2.1 Blocks Occupation Data

The Montréal subway system uses a fixed block signalling system with fixed blocks of varying lengths to transmit speed instructions to trains. The autopilot system (AP) of the Montréal subway acts as an Automatic Vehicle Location (AVL) system which informs the control centre of blocks occupancy status throughout the network. Each time an event occurs, a signal is sent to the control centre to display the position of the train. There are two types of events detected by the APs: (1) the entry of a train into a block and (2) the exit of the train from a block. The data transmitted to the control centre for each of these events include timestamp the event, block ID, line number, type of the event and ID of the train having the event. Table 3.2 describes the contents of the raw blocks data.

Table 3.2 Structure of the block occupation data

Field	**Content**	**Domain**
Ligne	Line number	1, 2, 4 ,5
EventDate	Timestamp of the event	(yyyy-mm-dd hh:mm:ss)
Etat	Statues of the event	(OCCU or LIBR)
NumeroTournee	Id of the train	1 to 73
ObjectNameText	Id of the block	1W129, 5W88, …

In Table 3.2, Id of the train is a unique identifier for each train operating in the system. Etat is either OCCU (occupied) or LIBR (Left), and ObjectNameText is the unique identifier for each block on a rail track. Table 3.3 shows a sample of the block raw data as an example.

Table 3.3 A sample of the block occupation raw data

index	Ligne	EventDate	Numero Tournee	Etat	ObjectNameText
0	1	2018-01-01 5:00:08	64	LIBR	1S42-RE0105-CT-CDV-GEN-1W14
1	1	2018-01-01 5:00:13	61	LIBR	1S44-LT0123-CT-CDV-GEN-1W5
2	1	2018-01-01 5:00:13	61	OCCU	1S46-SC0475-CT-CDV-GEN-1W1
3	1	2018-01-01 5:00:24	61	LIBR	1S44-LT0123-CT-CDV-GEN-1W3
4	1	2018-01-01 5:00:26	61	OCCU	1S46-SC0475-CT-CDV-GEN-1E1
5	1	2018-01-01 5:00:29	64	OCCU	1S42-RE0105-CT-CDV-GEN-1W18
6	1	2018-01-01 5:00:41	61	LIBR	1S46-SC0475-CT-CDV-GEN-1W1
7	1	2018-01-01 5:00:43	64	LIBR	1S42-RE0105-CT-CDV-GEN-1W16
8	1	2018-01-01 5:00:45	64	OCCU	1S40-SR0127-CT-CDV-GEN-1W20
9	1	2018-01-01 5:00:54	64	LIBR	1S42-RE0105-CT-CDV-GEN-1W18
10	1	2018-01-01 5:00:54	64	OCCU	1S40-SR0127-CT-CDV-GEN-1W22

3.2.2 Incident Data

The second type of data used in this study is the incident logs which are the manual inspection records of subway incidents lasting 5 minutes or longer. This dataset includes information such as occurrence time, location, cause, and duration of disruptions, provided by the metro operator. Table 3.4 describes the contents of the raw incident log data.

Table 3.4 Structure of the Incident data

Field	**Content**	**Domain**
Date incident	Date of incident occurrence	yyyy-mm-dd
Heure de l'incident	Time of incident occurrence	hh:mm
Delai	Incident delay duration in minutes	Integers > 5
Lieu	Incident location	block code:station name
Ligne	Line where incident occurred	1, 2, 4, 5
Code grandes causes	Incident cause	94 different causes

Table 3.5 presents a sample of the raw incident logs data.

Table 3.5 Sample of incident logs data

Date	**Time**	**Location**	**Line**	**Cause**	**Delay (min)**
2018-01-01	03:05	Cartier	orange	Door failure	7
2018-01-01	03:08	Place-des-Arts	green	Nuisance	6
2018-01-01	05:40	St-Michel	blue	Human factors	10
2018-01-01	08:59	Guy-Concordia	green	Person on the track	8
2018-01-01	17:33	Beaudry	green	Police intervention	6

CHAPTER 4 METHODOLOGY

This chapter presents the methodological framework of the thesis. To reach the research objectives, a two-stage framework is designed as shown in Figure 4.1. The first stage is aimed at extracting train trajectories from block data. In this phase, after processing block data and obtaining the block sequence for each metro line, direction, trajectories are extracted at the station level for each day in the study period. The goal of the second stage is to measure incident impacts using incident data and trajectory data. In this stage, the station occupancy time is selected as an indicator for quantifying train delays. After computing station delays for all trajectories, incidents are then identified in trajectory data by cross-referencing with incident logs. Subsequently, several impact indicators are defined to quantify incident effects on operation and finally, a clustering method is used to identify severity patterns of incidents. Details of these analysis steps are explained in the following sections.

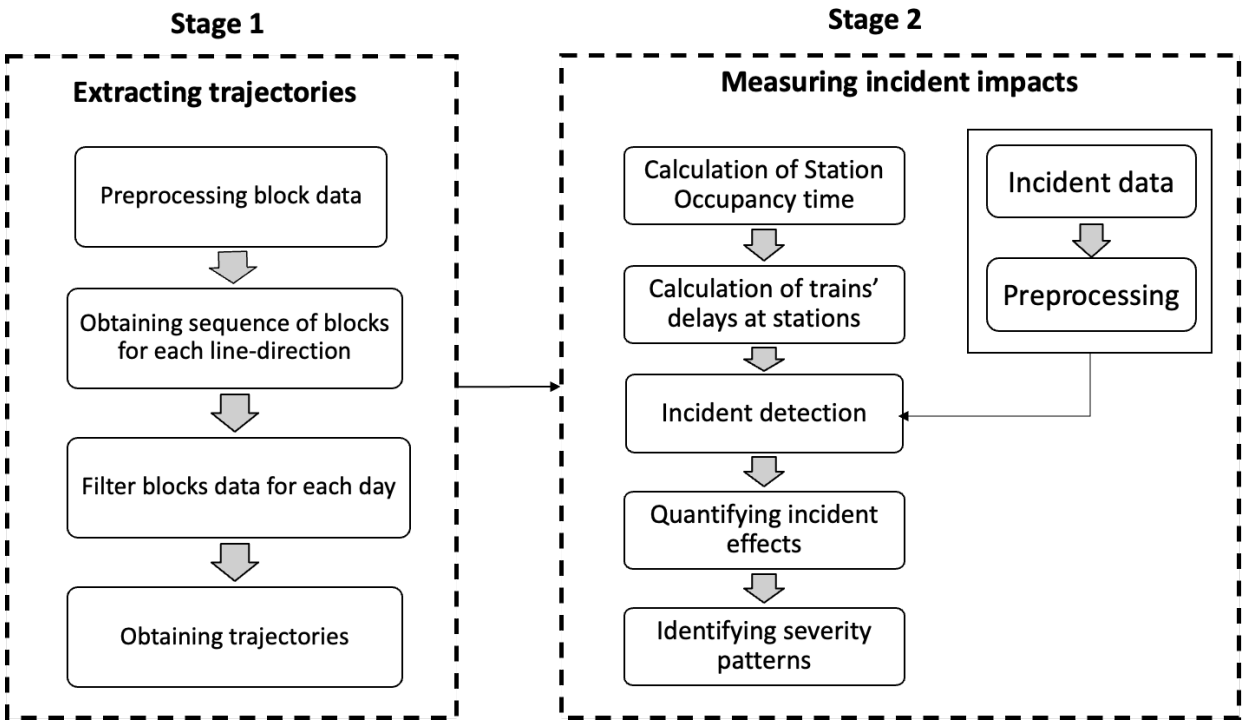


Figure 4.1 Methodological framework of the thesis

4.1 Stage 1: Extracting Trajectories

In this section, we begin by outlining the preprocessing steps for the blocks data. We then provide a detailed explanation of the process for retrieving trajectories from the block data.

4.1.1 Preprocessing Block Data

The raw operational data lacks distinction between station blocks and between-station blocks, making it unclear which block corresponds to which part of the track. Additionally, the direction of the train route is not discernible from the data. Thus, the first step is to enrich this data with more information including the direction of train movement, block length, and stations. To achieve this, the geometry of the metro lines is recreated using the structural map of the subway network. This map contains detailed information about the blocks for each metro line and direction. A part of this map is shown in Figure 4.2. For instance, block id *1W108* corresponds to station *Jolicoeur* for the west direction of line 1.

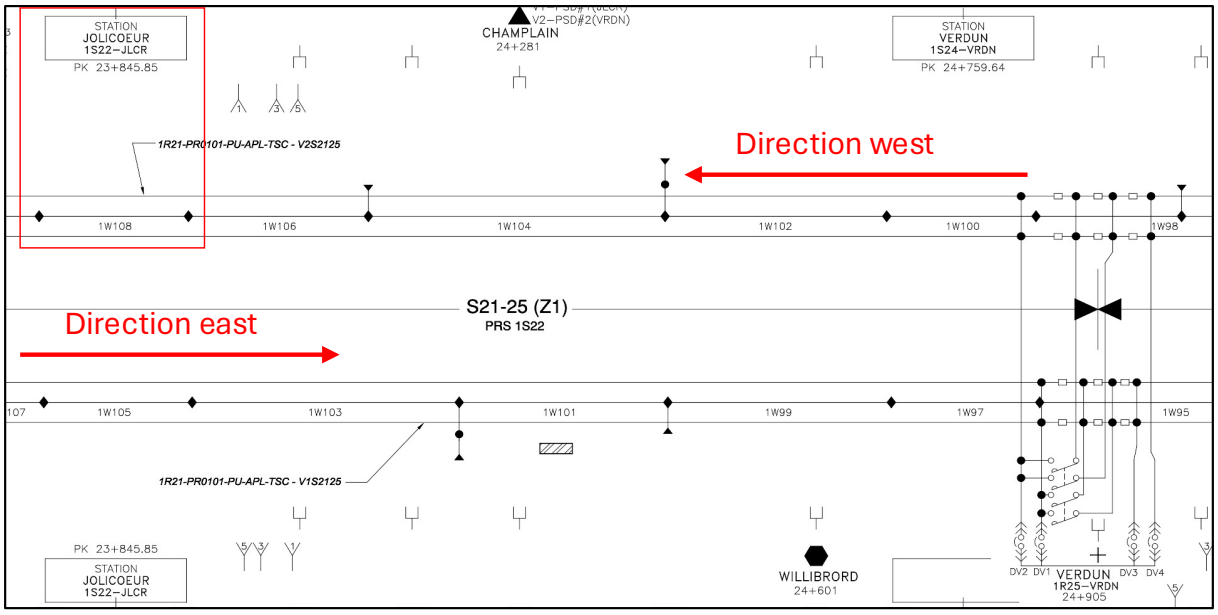


Figure 4.2 A part of the structural map of the Montréal subway

After extracting all the necessary information regarding each block, a new dataset is generated called as block sequence. Table 4.1 shows the content of the block sequence data for a part of the west direction of the line 5. According to this table, if a block corresponds to a station, it has the name of a station in the station column; otherwise, it has the value 0. The column length shows the

length of the block in metres. If the direction is the east, the direction value equals 1, otherwise it's 2.

Table 4.1 Block sequence data

Line	Direction	block	length	station
5	2	5W50	337	0
5	2	5W52	216	0
5	2	5W54	160	OUTREMONT
5	2	5W56	204	0
5	2	5W58	365	0
5	2	5W60	244	0
5	2	5W62	269	0
5	2	5W64	160	EDOUARD MONTPETIT
5	2	5W66	242	0
5	2	5W68	229	0
5	2	5W70	189	0
5	2	5W72	160	UNIVERSITE DE MONTREAL

In the next step, the new features presented in Table 4.1 were added to the original dataset based on the column *block*. It should be noted that in the preprocessing step, the data related to service hours are considered. Service hours are the time of days when the subway system is open to passengers. For the case of Montréal subway, service hours are between 5:30 a.m. and 2:00 a.m. in the next day with a little variation on weekends. Also, all the blocks data that are outside of the normal traffic lanes are removed. In other words, blocks located outside the terminal stations (i.e., blocks placed in garages) are removed, since those data are not related to the purpose of this study.

4.1.2 Retrieving Trajectories

Now that the blocks and stations of each line and direction are identified, it is possible to retrieve trains' trajectories. To follow the passage of each train on the network, the following steps are taken:

1. For each day, filter the data for each line and direction and sort them by datetime.
2. Get the list of trains (unique values) and sequence of stations (based on the line and direction) for the filtered data in step 1.
3. Filter data for each train (only the data for station blocks).

4. Add a new column to the data to name the trajectory number.
5. Start from trajectory number 1. Loop over blocks for each train considering chronological order of stations (start from the first station to the last one) and put trajectory number in the column trajectory.
6. Update the trajectory number (increase 1 step) if a train reaches the starting station of its line and direction.
7. Once all trajectories for each train have been identified, proceed to renumber the trajectories sequentially based on their time order, starting from trajectory 1 and continuing until the last trajectory of the day.

Table 4.2 provides an example of the transformed trajectory data following the described steps. The column *traject* shows trajectory number corresponding to train ID (in this example train 7) while the column trajectory shows the trajectory number of the day. To clarify, the first trajectory associated with train ID 7 corresponds to the 20th trajectory of the day.

Table 4.2 An Example of the Transformed Trajectory Data

DateTime	train	line	direc	stat	station	traject	trajectory
2018-01-03 07:06:26.009	7.0	1.0	1.0	O	ANGRIGNON	1	20.0
2018-01-03 07:08:21.499	7.0	1.0	1.0	L	ANGRIGNON	1	20.0
2018-01-03 07:09:05.280	7.0	1.0	1.0	O	MONK	1	20.0
2018-01-03 07:09:58.075	7.0	1.0	1.0	L	MONK	1	20.0
2018-01-03 07:10:50.383	7.0	1.0	1.0	O	JOLICOEUR	1	20.0
2018-01-03 07:11:41.178	7.0	1.0	1.0	L	JOLICOEUR	1	20.0
2018-01-03 07:12:14.500	7.0	1.0	1.0	O	Verdun	1	20.0
2018-01-03 07:13:04.995	7.0	1.0	1.0	L	Verdun	1	20.0
2018-01-03 07:13:28.371	7.0	1.0	1.0	O	De L'Église	1	20.0
2018-01-03 07:14:18.271	7.0	1.0	1.0	L	De L'Église	1	20.0

A useful way of examining the operation of railways is to visualize the trajectories on a space-time (station-time) diagram. Figure 4.3 shows the trajectories of trains for the green line of the Montréal

subway during the evening hours of a weekday in January 2018. In this graph, the x-axis shows the time and y-axis shows the sequence of stations of the line. The flat parts of trajectories indicate time spent at stations while diagonal parts show the movement of trains between stations.

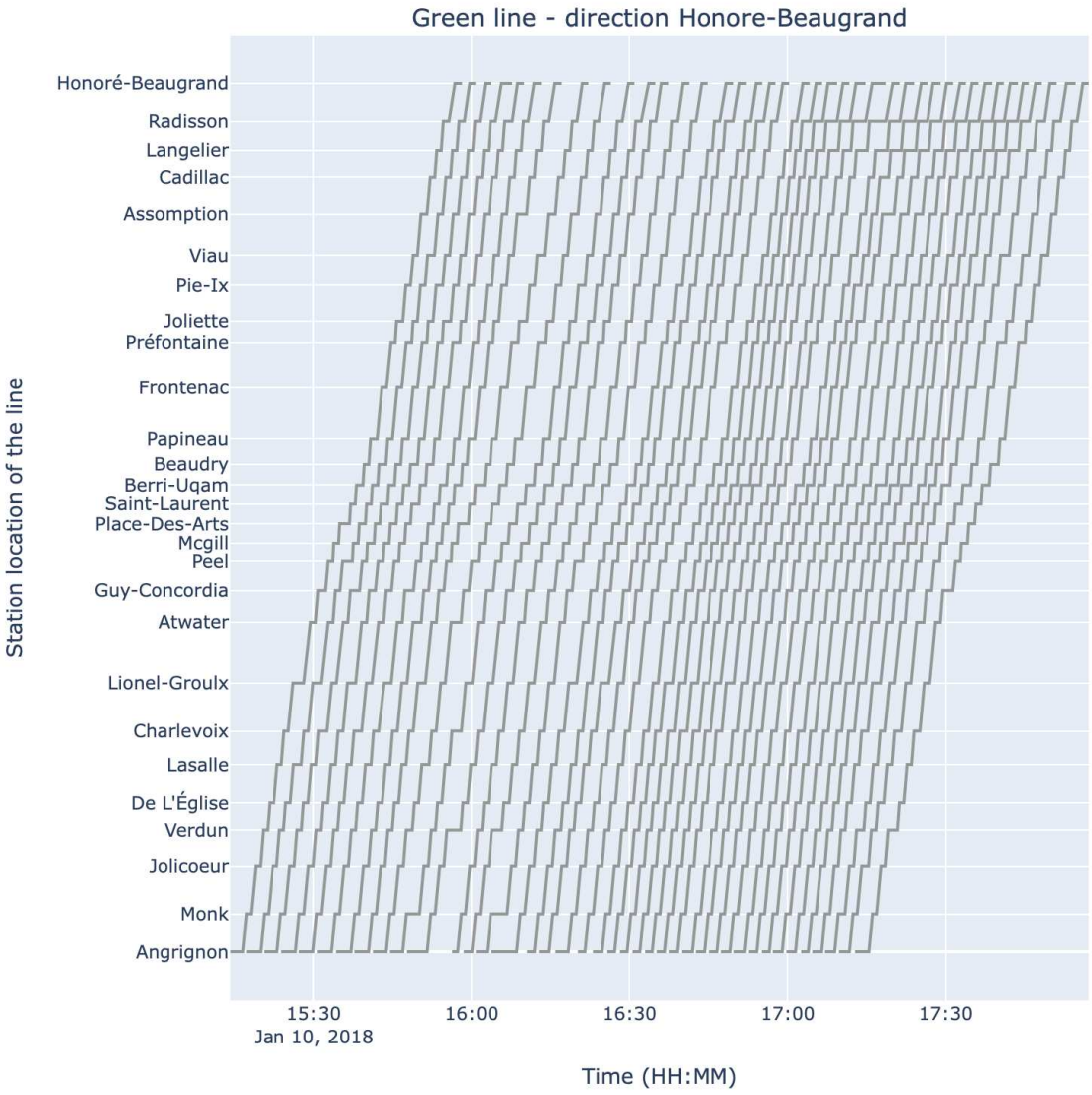


Figure 4.3 Space-time diagram of train trajectories for the green line on Jan 10, 2018

4.2 Stage 2: Measuring Incident Impacts

As mentioned previously, the main objective of this study is to quantify the impacts of incidents on subway operation. Stage 2 of this methodology is developed to achieve this vital goal through a multi-step analysis. In the first step, we propose to calculate the station occupancy time (SOT)

for all stations of each trajectory and then station delays are quantified accordingly. Next, by merging incident logs with train trajectories, spatiotemporal propagation of service disruptions is identified. Subsequently, various incident severity measures are defined to capture the impact of incidents on operation. Finally, K-means clustering method is applied to identify severity patterns of the Montréal subway incidents.

4.2.1 Step 1: Calculation of Station Occupancy Time

In trajectories data derived from stage 1, timestamps of train movements across track sections are recorded. As mentioned in section 2.1 the Station Occupancy Time (SOT) refers to the time difference between the arrival and departure of a train from a station. Thus, SOT can be easily calculated by subtracting the activation time from the deactivation time of a station block.

Let *t* be the trajectory number of line *l* in direction *d* on a service day. SOT of trajectory t for station s can be calculated by Equation 1:

$$SOT_{tlds} = DT_{tlds} - AT_{tlds} \tag{4.1}$$

Where:

DT_{tlds} *and* AT_{tlds} *stand for deactivation time and activation time of the block s, line l, and direction d, respectively.*

4.2.2 Step 2: Quantifying Train Delays

In subway systems, the main objective is to maintain a planned service frequency and train timetables are difficult to maintain. Thus, a deviation of the train's location from a pre-planned timetable might not indicate a delay (the service is rather based on frequency than schedule). In the Montréal metro system, when incidents occur, a standard operational procedure is to halt all trains operating on the affected track at the next stations. Consequently, this precautionary measure leads to knock-on delays appeared in the station occupancy time (SOT) for each affected train. Thus, in this case, SOT is considered as an indicator to quantify delays resulting from incidents.

Under normal operating conditions, SOT at each station typically falls within a specific time range. However, during incidents, SOT significantly increases as trains are halted at stations until the

incident is resolved. To better illustrate the difference between the normal and delayed operation, an example of trajectories is presented in Figure 4.4. This figure displays the space-time diagram of the green line operation during a 3-hour period between 15:00 and 18:00 on January 11, 2018. It is evident that from 15:00 to around 16:20, service was running under normal operation (little variation of SOT at different stations). Around 16:25, service was interrupted at station Beaudry (sudden increase of SOT) which led to the increase of SOT of other trains at the preceding and succeeding stations. After the incident was resolved (around 16:45), the operation returned to normal condition and SOT of all trajectories remained at a stable level.

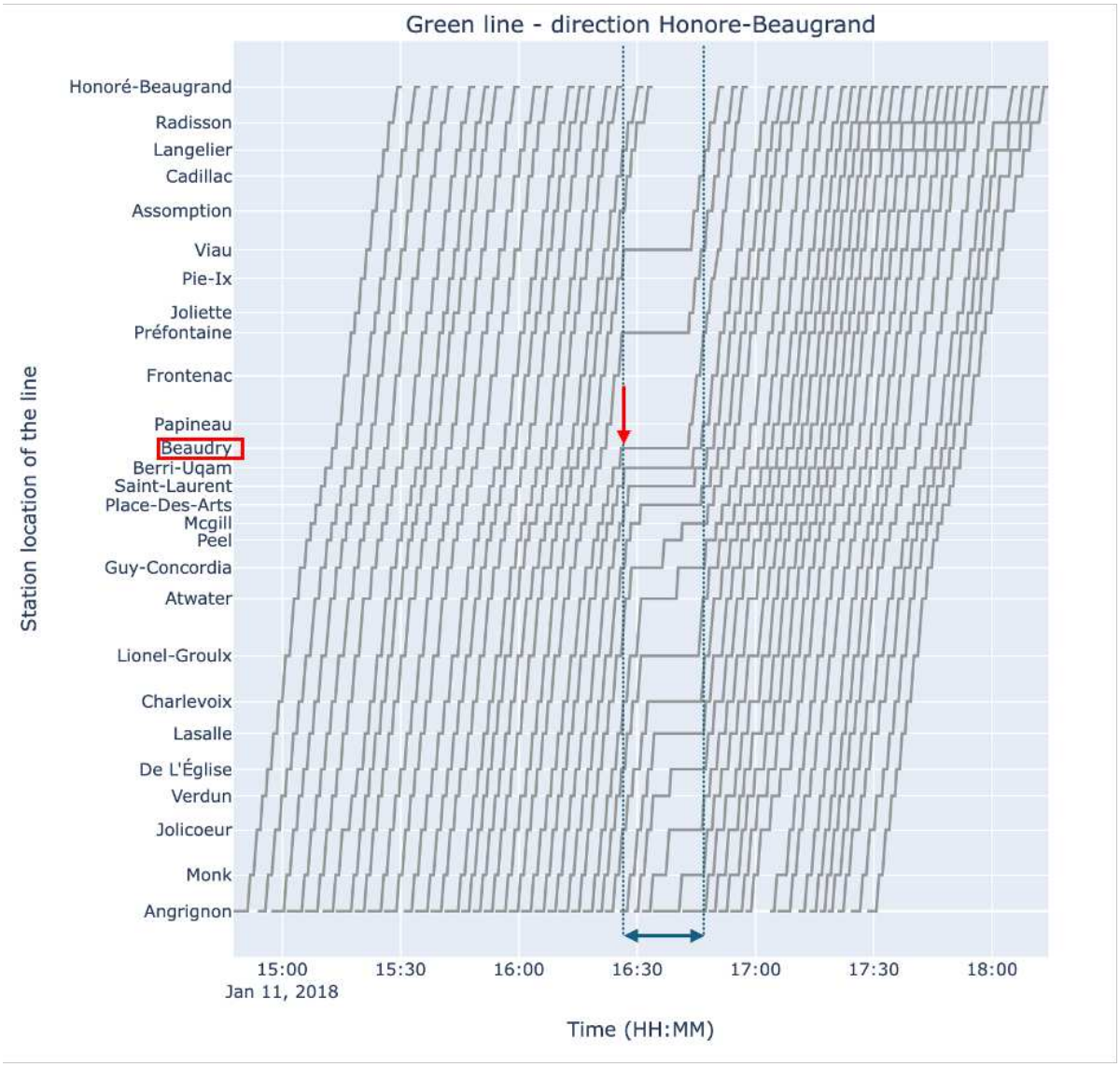


Figure 4.4 Station-time diagram for green line trains on January 11, 2018, with an incident occurring at 16:25 at station Beaudry

Figure 4.4 clearly shows that incident occurrence results in an increase of SOT, and it can be used as a reliable indicator for quantifying train delays at stations. Thus, for each trajectory the difference between observed SOT and the median SOT for the time of trajectory is calculated and considered as the delay time. The median was chosen because the distribution of SOT for all stations is right skewed due to service disruptions. To account for peak hours and the variation of passenger demands, the median SOT is calculated for each station during different hours on weekdays and weekends each month. It should be noted that the reported delays in the incident dataset are on the minute scale, and the delays equal to or greater than 1 minute are extracted from trajectory data for further analysis. The pseudo-code for the implementation of step 2 is provided in Figure 4.5.

```
1.  Input: trajectories data for each month
2.  LD = list of all the pairs of metro line-direction
3.  S = list of all the stations for each line-direction in LD
4.  H = list of service hours
5.  SOT = Station Occupancy Time

6.  For each LD:
7.      For each station in S:
8.          For each hour in H:
9.              Calculate the median of SOT over weekdays and store it in the
                column median SOT for all corresponding rows.

10.             Calculate the median of SOT over weekends and store it in the
                column median SOT for all the corresponding rows.
11.         End loop for service hours
12.     End loop for stations
13. End loop for line-direction pairs

14. Subtract column SOT from the column median SOT.

15. Output: trajectories data with a new column representing delay time at stations
```

Figure 4.5 Pseudo-code for quantifying train delays

4.2.3 Step 3: Incident Identification in Trajectory data

Detection of incidents in trajectory data using incident logs data is a crucial step for identifying the spatiotemporal propagation of incident delay. In the incident dataset, the time, line, station, delay duration and the cause of the incident are registered. The first step is to process these data so that it can be integrated with trajectory data. New features including year, month, day of the month and time-minute (time of incident occurrence in minute), are added to incident logs data. The incident delay reported in incident logs is what we define as the primary delay or the source of delay. To identify primary delays in trajectory data, the following steps are taken:

1) Get the year, month, day, time-minute and line number of each incident record.
2) Look into the corresponding trajectories data and label the closest matched row as the primary delay.

To address potential inaccuracies in manually recorded incident data, a buffer time *(T)* is incorporated for incident time occurrences. For instance, if an incident is reported at time *t* on line L, the algorithm examines the trajectories of line L within a time range of $(t - T)$ to $(t + T)$. The algorithm then labels the closest matched row, considering both time and location, as the source of the incident. After testing different values for *T*, it is determined that setting *T* to 10 minutes effectively identifies all incidents in the trajectories data.

To further assess the inferred incident and validate the dataset, the registered station and delay time of the incident are appended to the trajectories data. This process enhances the reliability and consistency of the incident information in the overall analysis. An example of incident identification in trajectory data is shown in Figure 4.6.

Incident data

Date	Time	Location	Line	Type	Delay
2018-01-11	16:26	beaudry	1	Blocage	16.0

arrival_time	trajectory	station	line	direc	SOT	Median SOT	Delay(sec)	primary_delay
2018-01-11 16:05:01.907000	125.0	angrignon	1.0	2.0	195.089	89.3925	105	0
2018-01-11 16:15:52.755000	132.0	guy-concordia	1.0	2.0	169.285	59.5450	109	0
2018-01-11 16:17:21.946000	128.0	angrignon	1.0	2.0	186.585	89.3925	97	0
2018-01-11 16:25:54.255000	134.0	beaudry	1.0	1.0	1024.210	54.9975	969	1
2018-01-11 16:26:11.038000	133.0	préfontaine	1.0	1.0	1014.310	53.2950	961	0
2018-01-11 16:26:36.625000	132.0	viau	1.0	1.0	1027.610	55.0950	972	0
2018-01-11 16:26:44.496000	135.0	berri-uqam	1.0	1.0	1037.609	76.6425	960	0
2018-01-11 16:27:51.756000	136.0	saint-laurent	1.0	1.0	1000.214	56.3950	943	0

Figure 4.6 An example of incident detection in trajectories using incident log data

4.2.4 Step 4: Quantifying Incident Effects

Identification of Secondary Delays

Data resulted from step 3 includes delay information for trains, with certain rows labelled as primary delay. In this study, delays are categorized into two types: primary delay (PD) and knock-on or secondary delay (SD). PDs refer to delays caused by incidents, whereas SDs are delays resulting from the propagation of a PD to either upstream or downstream stations along a metro line. The steps for detecting secondary delays are summarized as follows:

1) Arrange all delay data based on the arrival time of the trains at stations.
2) For each row labelled as an incident, set the arrival and departure time of that train as the starting and ending time of the incident, respectively.
3) Extract the sequence of rows where the line number matches the line of the incident, and train arrival time falls within the starting and ending time of the incident. Mark all these rows as secondary delays.

An example of a PD and its consequent SDs is shown in Table 4.3. The incident occurred for trajectory 134 (PD=1) and it affects trajectories in both directions (SD column equal to 1 means that train is affected).

Table 4.3 An example of primary delay and its resulting secondary delays

arrival_time	trajectory	station	line	direc	Delay(sec)	primary_delay	secondary_delay
2018-01-11 16:25:54.255000	134.0	beaudry	1.0	1.0	969	1	0
2018-01-11 16:26:11.038000	133.0	préfontaine	1.0	1.0	961	0	1
2018-01-11 16:26:36.625000	132.0	viau	1.0	1.0	972	0	1
2018-01-11 16:26:44.496000	135.0	berri-uqam	1.0	1.0	960	0	1
2018-01-11 16:27:51.756000	136.0	saint-laurent	1.0	1.0	943	0	1
2018-01-11 16:28:24.118000	137.0	mcgill	1.0	1.0	73	0	1
2018-01-11 16:28:32.690000	138.0	guy-concordia	1.0	1.0	411	0	1
2018-01-11 16:29:19.781000	131.0	angrignon	1.0	2.0	75	0	1
2018-01-11 16:30:52.291000	137.0	place-des-arts	1.0	1.0	858	0	1
2018-01-11 16:30:59.098000	140.0	lionel-groulx	1.0	1.0	806	0	1
2018-01-11 16:31:00.636000	139.0	atwater	1.0	1.0	480	0	1

A space-time diagram can better illustrate delay propagation of an incident. The example of Table 4.3 is presented in the Figure 4.7. In this diagram, the red line shows the trajectory with primary delay where the incident occurred. It is evident that incident delay has propagated rapidly across the line and affected trajectories in both directions.

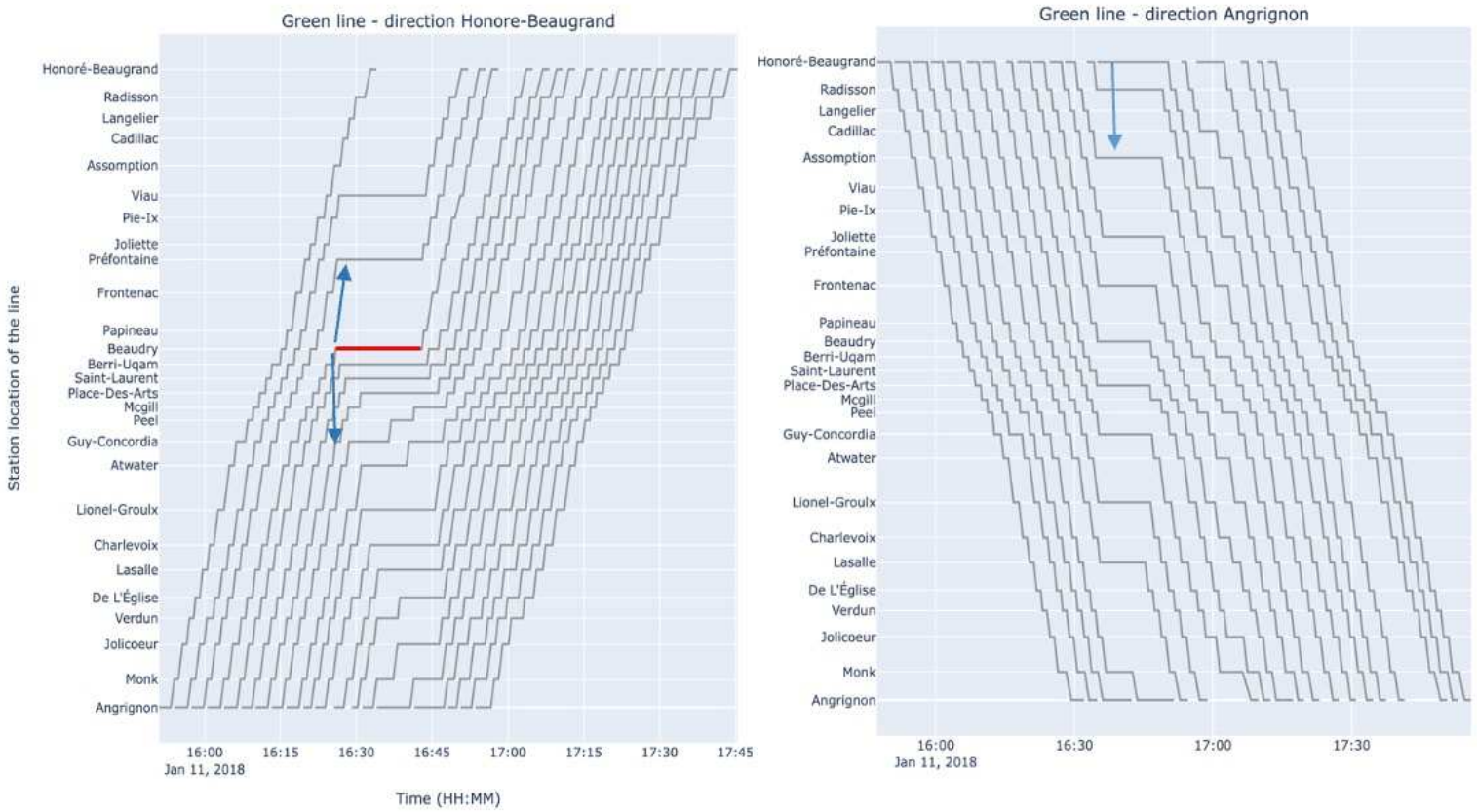


Figure 4.7 Station-time diagram for green line trains on January 11, 2018, highlighting an incident and its secondary delays in both direction

Defining Incident Impact Indicators

After labelling all the secondary delays for all incidents, it is possible to identify the influence of each incident on the operation. Four indicators are defined as shown in Table 4.4. to capture the overall impact of incidents on a metro line. These indicators are used as the input of the next step to reveal the propagation patterns of different incidents.

Table 4.4 Incident impact indicators

Indicator	**Detail**
Number of affected trains (NAT)	Total number of unique train IDs that were delayed due to an incident.
Number of affected train-stations (NATS)	Total number of unique train-stations pairs that were delayed due to a PD.
Total delay time (TDT)	Sum of all the delays caused by an incident. TDT = PD + sum (SDs)
Both directions affected (BD)	Takes value 1 if there are delayed trains in both directions, 0 otherwise.

For instance, for the incident shown in Figure 4.7, affected trajectories are in both direction (BD=1), NAT is 32 and NATS is 52 (several trains were delayed at multiple stations). Total delay time caused by this incident is approximately 350 minutes.

Table 4.5 shows a sample of incident data with their impact indicators.

Table 4.5 Sample of incident data with their impact indicators

Cause	Station	line	hour	NAT	NATS	BD	Primary delay (min)	Total delay (min)
Tunnel blockage	Peel	1	9	9	11	1	8.30	36.77
Assault	Cadillac	1	17	8	8	1	7.98	58.43
Tunnel blockage	Villa-Maria	2	18	10	18	1	19.85	156.80
Smoke	Frontenac	1	9	9	10	1	5.58	25.60

4.2.5 Step 5: Identifying Severity Patterns

In this step, a clustering method is applied to identify severity patterns of different incident types. The goal of clustering is to reveal inherent groupings or clusters within a given set of patterns, points, or objects. Clustering algorithms are broadly categorized into hierarchical and partitional methods (Jain, 2010). Hierarchical clustering recursively finds nested clusters by splitting or merging clusters, while partitional algorithms find all the clusters simultaneously as a partition of the data (Cerreto et al., 2018; Jain, 2010). K-means is the simplest and the most popular partitional clustering algorithm that has strong robustness on high-dimensional and multicollinear datasets. Let $X = \{x_i, i = 1, \dots, n\}$ represent a set of n-dimensional points to be clustered into K clusters denoted by $C = \{C_k, k = 1, .., K\}$. The K-means algorithm aims to find a partition that minimizes the squared error between the empirical mean of a cluster and the points within that cluster. Denoting the mean of cluster C_k as μ_k, the minimization of the sum of the squared error over all K clusters is defined as Equation 2:

$$\min\left(\sum_{k=1}^{K}\sum_{x_i \in Ck}\left||x_i - \mu_k|\right|^2\right) \tag{4.2}$$

The main steps of K-means algorithm are as follows:

1) Choose the number of clusters (K) and initiate cluster centroids.
2) Assigning every observation to the cluster with the closest centroid.

3) Calculating the new centroid of each cluster as the mean of the observations within the cluster.
4) Repeat steps 2 and 3 until the cluster membership stabilizes.

The input of clustering is the impact indicators derived from the previous step, including NAT, NATS, BD and TDT. As clustering algorithms like K-means are distance-based, standardization of features is necessary to ensure equal contribution to distance calculations (Jain, 2010). Without standardization, features with larger scales may disproportionately influence clustering outcomes. Standardization involves centering the features (subtracting the mean from each data point) and then scaling them by the standard deviation, resulting in data with a mean of 0 and a standard deviation of 1. This process can be represented by Equation 4.3:

$$x' = \frac{x - \mu}{\sigma} \tag{4.3}$$

Where x' represents the standardized feature, x denotes the original feature value, μ is the mean of the feature, and σ is the standard deviation of the feature.

Table 4.6 presents a sample of standardized features used as input for clustering. Rows correspond to incidents, while columns represent normalized values for features NAT, NATS, BD, and TDT.

Table 4.6 Sample of clustering input

Incident	NAT	NATS	TDT	BD
1	-0.622	-0.533	-0.395	0.305
2	-0.746	-0.706	-0.245	0.305
3	-0.498	-0.129	0.433	0.305
4	-0.622	-0.591	-0.472	0.305
5	1.235	1.083	0.374	0.305

CHAPTER 5 RESULTS

This chapter presents the results of this research. It starts with descriptive statistics of incident dataset followed by an analysis of incident impact indicators. A comparative examination between identified incident delays and reported ones is then presented. Lastly, a detailed analysis of clustering results for incident severity patterns is provided.

5.1 Descriptive Statistics of Incident Data

5.1.1 Temporal Analysis

In the incident logs, a total of 983 incidents were recorded in the year 2018, excluding the yellow line. The frequency distribution of incident delay is displayed in Figure 5.1. Incident delay has a right-skewed, heavy-tailed distribution, meaning short delays occurring more frequently than long-lasting delays. Overall, 75% of the incident delays were 10 minutes or less and only 5% of incidents had a delay of more than 30 minutes. On average, incidents caused a delay of 10.55 minutes, with a standard deviation of 11.62.

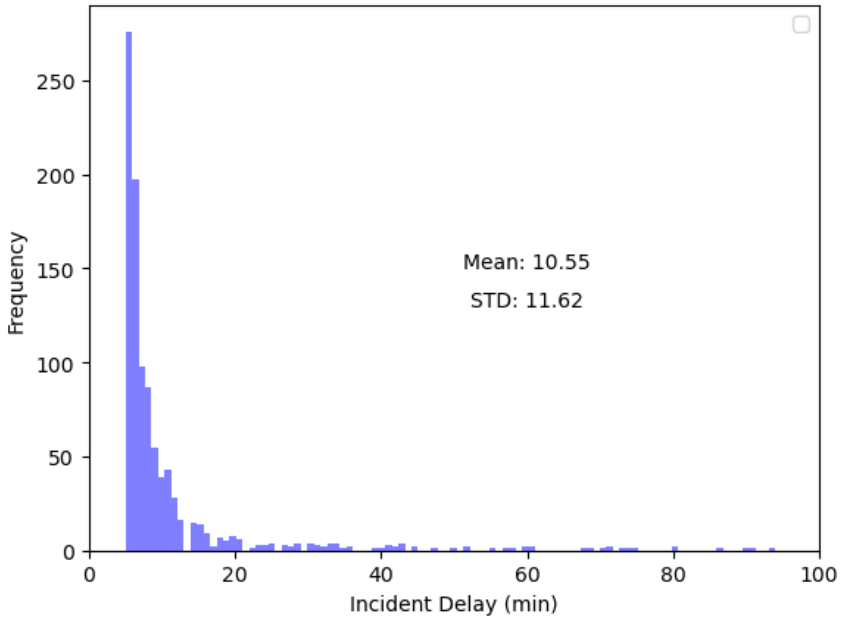


Figure 5.1 Frequency distribution of incident delays

To know how incidents are temporally distributed, the distribution of incident occurrence across service hours are shown in Figure 5.2. According to this figure, two distinct time intervals for incident delays can be distinguished:

Off-peak time : due to low frequency of service and lower density of travel demands during these periods, train operations are less vulnerable to incidents and as a result, fewer incidents occur.

Peak time (typically from 7 to 9 in the morning and from 15 to 18 in the evening) : in this period, the number of incidents increases suddenly (around 7 and 16) and remains relatively high. The increased travel demands and higher frequency of trains during peak hours likely contribute to the larger number of incident occurrence.

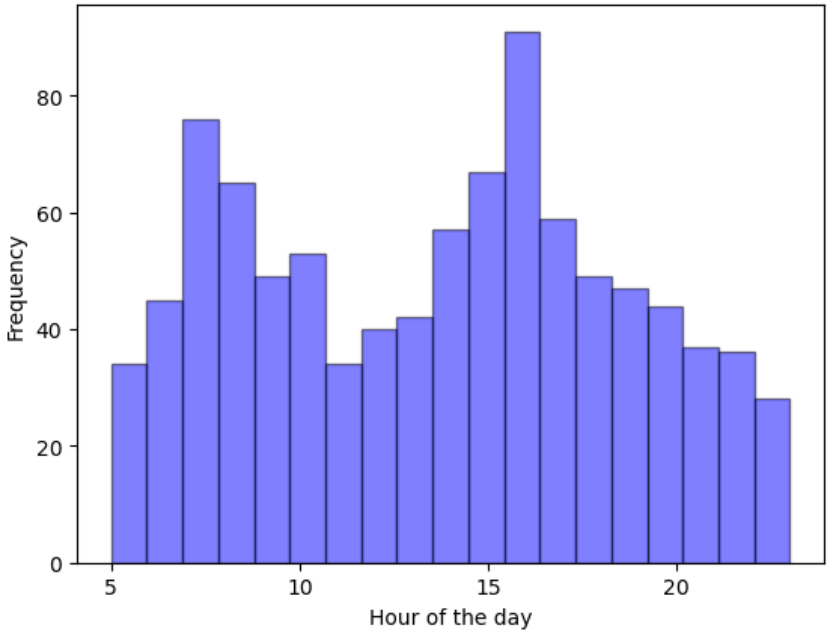


Figure 5.2 Frequency distribution of incident occurrence across service hours

5.1.2 Spatial Analysis

Stations and sections operate under varying conditions, leading to different probabilities of incident occurrences. Figure 5.3 shows the spatial distribution of incident occurrences across the subway network, with dot sizes indicating the frequency of incidents. Remarkably, terminal stations across all three lines had the highest incident occurrences. This might be

due to their high passenger volume, infrastructure challenges, or specific vulnerabilities that need to be addressed. Interestingly, stations located away from congested urban areas experienced fewer incidents compared to those closer to the city centre. This can somehow illustrate the effect of passenger congestion on incident occurrence. Additionally, lines green and orange experienced significantly more incidents than line blue. These two lines together comprise around 89% of all the reported incidents. The extended routes and a higher number of stations make these lines more vulnerable to incident occurrences.

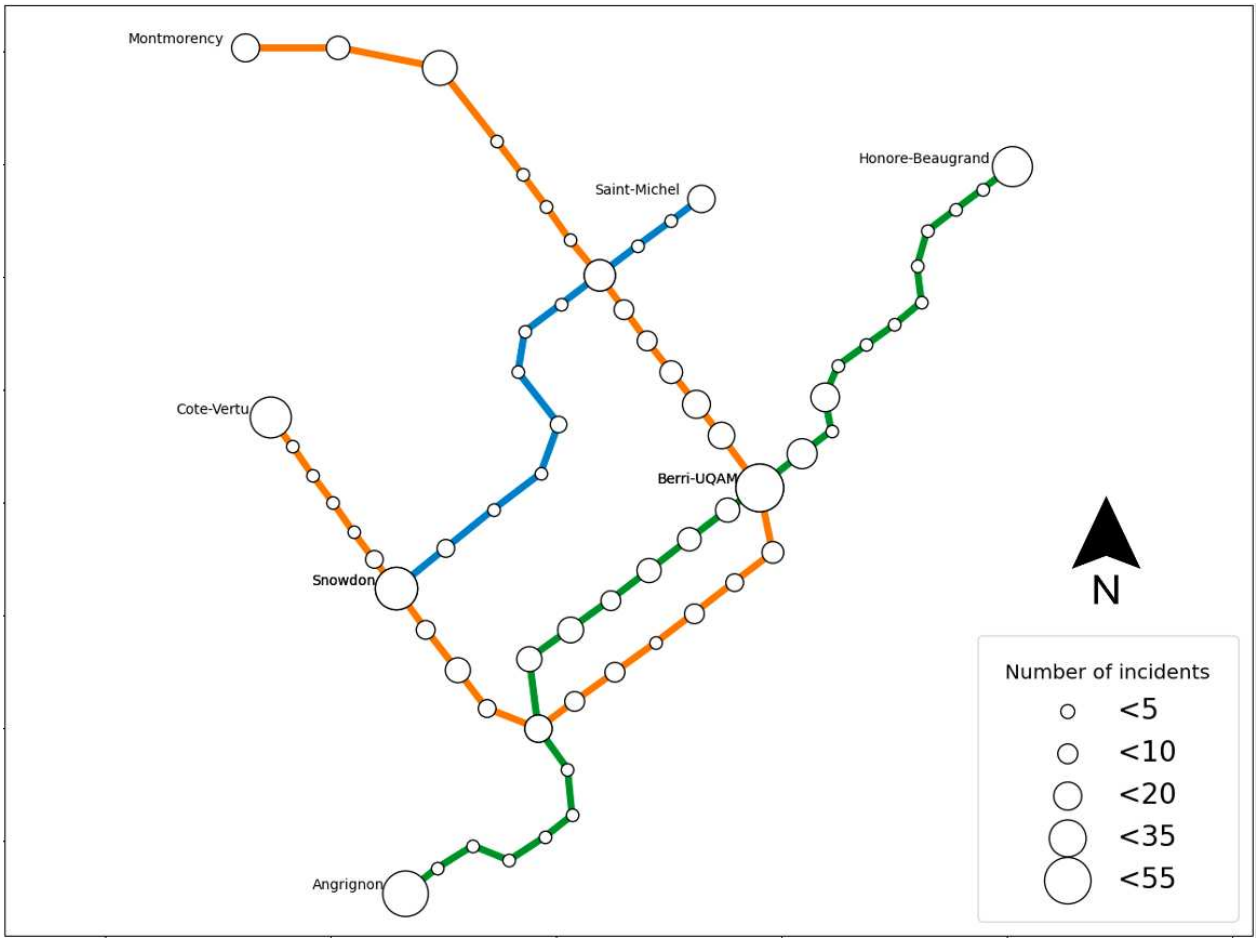


Figure 5.3 Spatial distribution of incident occurrences within the Montréal subway network

Table 5.1 presents the top 10 stations with the highest number of incident occurrences, among which 8 are either terminal or transfer stations.

Table 5.1 Stations with the most incident occurrences in 2018

Station	**Number of incidents**
Angrignon	53
Berri-Uqam	49
Snowdon	43
Côte-Vertu	42
Honorée-Beaugrand	41
Henri-Bourassa	31
Beaudry	21
Montmorency	20
Frontenac	19
S. Michel	19

5.2 Descriptive Statistics of Impact Indicators

As highlighted in the previous chapter, four indicators are defined to measure the impact of incidents on operation. In this section, a descriptive analysis of these impact indicators is presented.

5.2.1 Number of Affected Trains

The frequency distribution of the number of affected trains (NAT) by incidents is shown in Figure 5.4. The mean value for NAT is approximately 14, meaning each incident affected 14 trains on average. It should be noted that affected trains are those with a delay of 1 minute or more.

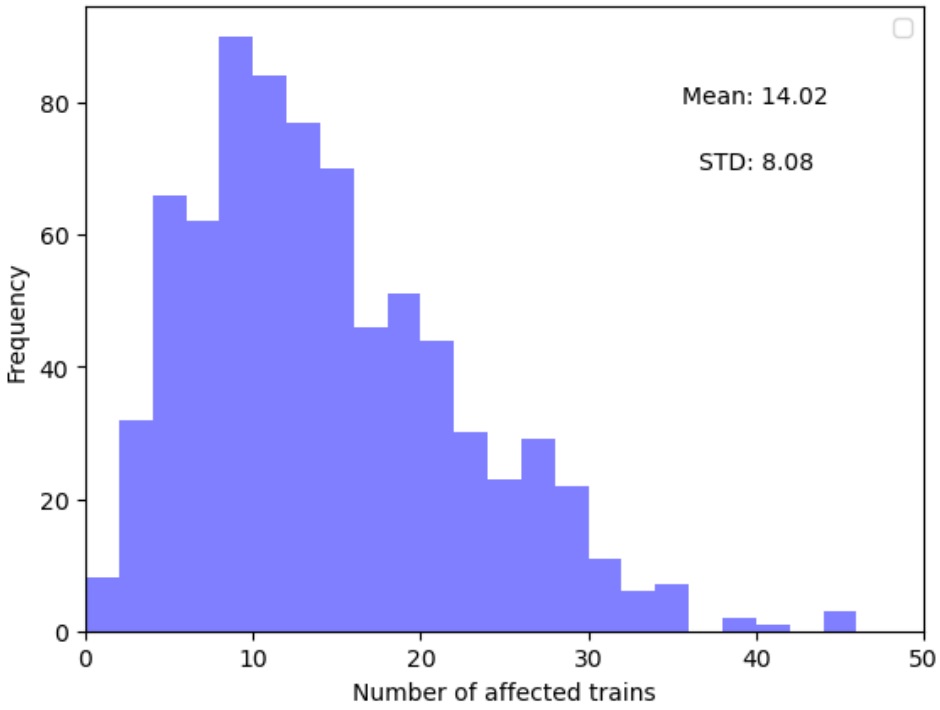


Figure 5.4 Frequency distribution of the number of affected trains

5.2.2 Number of Affected Train-station pairs

Another indicator for capturing an incident effect is the number of affected train-station pairs (NATS). The magnitude of this indicator is always equal or greater than the number of affected trains. The frequency distribution of NATS is shown in Figure 5.5. The mean value for NATS is approximately 20 with a standard deviation of 17.3.

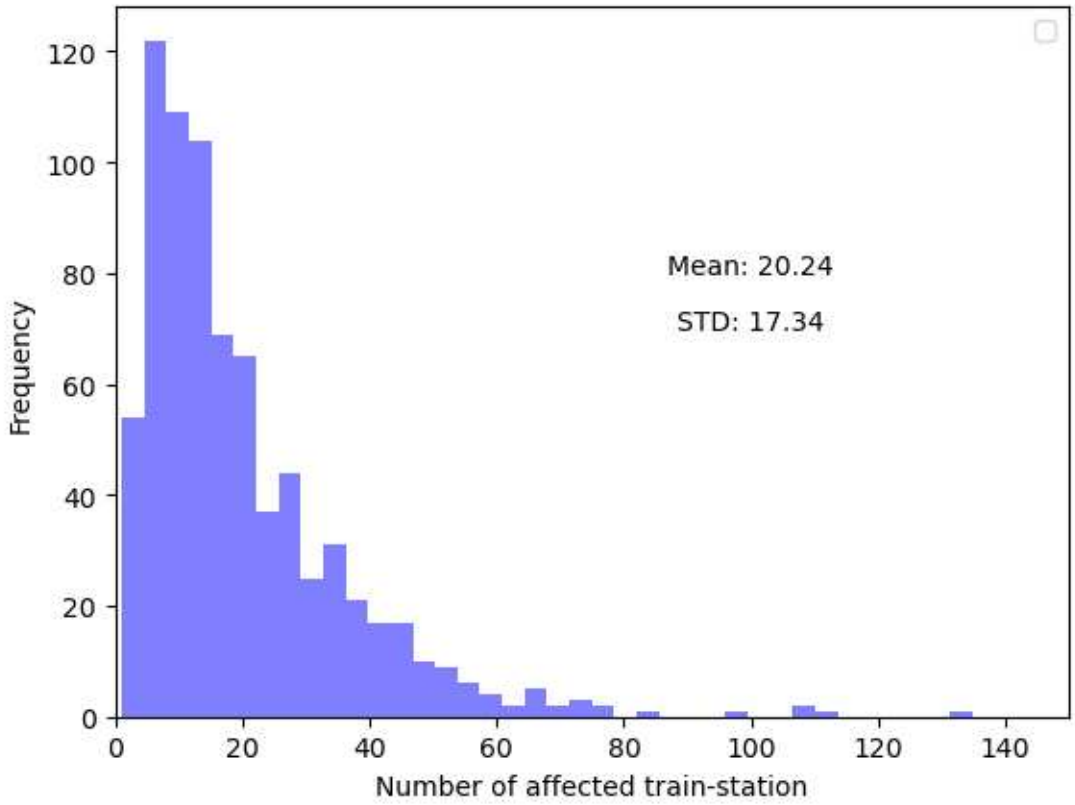


Figure 5.5 Frequency distribution of the number of affected train-station pairs

5.2.3 Total Delay Time

Figure 5.6 shows the frequency distribution of total delay time (TDT) caused by incidents in minutes. TDT has a highly skewed distribution, with an average of 94 minutes and a standard deviation of 145. The majority of incidents result in a TDT of less than 200 minutes, with only 10% of all incidents exceeding this threshold.

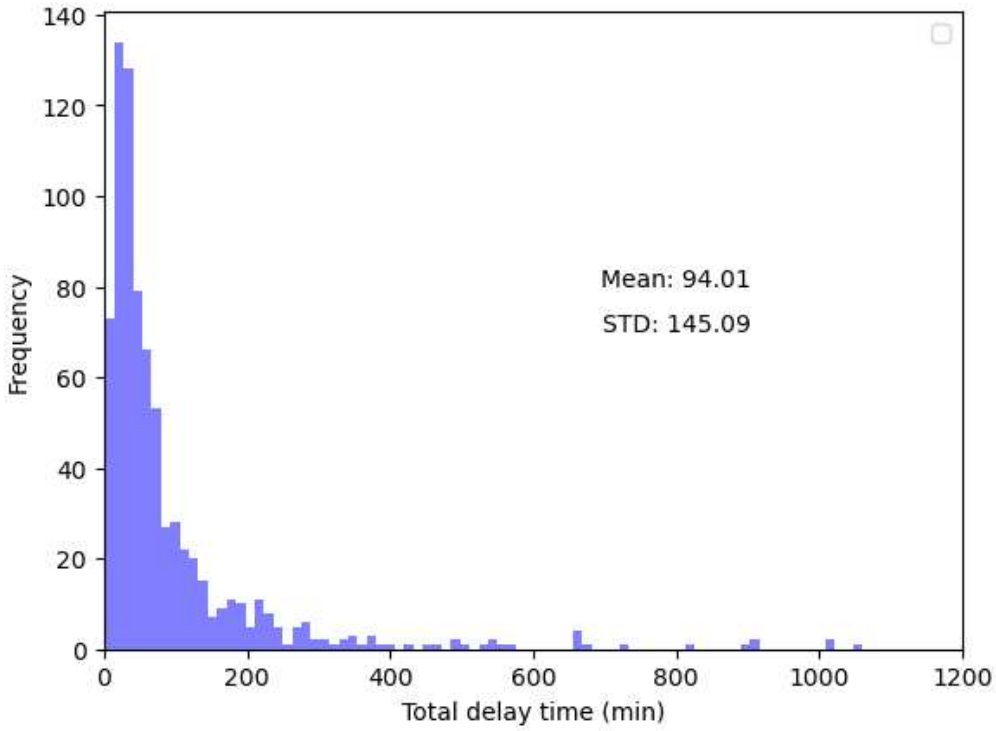


Figure 5.6 Frequency distribution of total delay time

5.3 Comparison of Identified Trajectories Delays with Reported Incidents Delays

As highlighted in chapter 2, incident logs are manually reported data that requires validation. The validation process involves comparing the incident delay time reported in the incident logs with the corresponding PD time identified in the operational data. In total, 765 incidents were identified in trajectory data based on incident log reports. The outcomes of this comparative analysis are shown in Figure 5.7. The x-axis represents the reported primary delay, while the y-axis displays identified primary delay. The red dashed line, indicating a 45-degree angle, serves as a reference. Proximity to this line indicates a minimal difference between reported and identified primary delays. Examination of Figure 5.7 reveals an overestimation of reported delays compared to identified delays, as most data points reside on the right side of the dashed line. Specifically, for primary delays under 20 minutes, identified delays are mainly smaller than the reported ones, suggesting a potential bias towards overestimating primary delays within the incident logs data.

Figure 5.7 highlights an important insight: relying solely on incident logs data may not yield accurate results, particularly for small incident delays. For instance, a 1-minute delay difference

for a 5-minute primary delay constitutes a 20% error, underscoring the potential impact on modelling outcomes in studies focused on incident delay prediction. This emphasizes the importance of considering and validating data from multiple sources to enhance the precision and reliability of predictive models, especially in scenarios where small delays carry substantial significance.

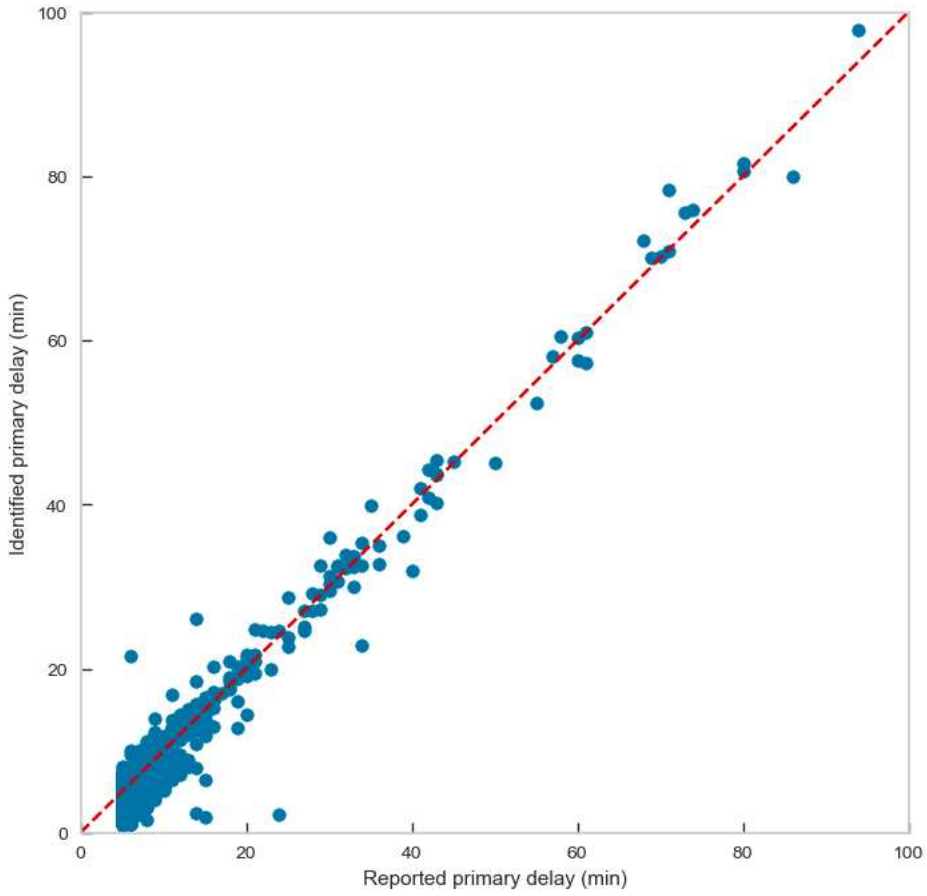


Figure 5.7 Scatter plot of the identified vs. reported primary delay

As mentioned in Chapter 3, incident logs from the Montréal subway are the documented records of the incidents lasting 5 minutes or more. To assess the completeness of this data, we compared it with the quantified delays derived from trajectory data. Upon examining our processed delay dataset, we discovered records indicating delays exceeding 5 minutes that were not labelled as primary or secondary delays, and therefore not linked to any incident record. These delays may represent incidents that went unrecorded in incident logs or could simply be delays resulting from

events or passenger congestion. This underscores the limitations of incident logs, which are prone to human errors and constrained in their monitoring range.

Overall, we identified 395 unlabelled records with delays exceeding 5 minutes during the study period. The distribution of these unlabelled delays is depicted in the Figure 5.8. Most of these delays were less than 10 minutes, with an average delay of 10.57 minutes and a median delay of 6 minutes. The proximity of the median value to the threshold set by the operator (5 minutes) implies potential human error in documenting smaller incidents.

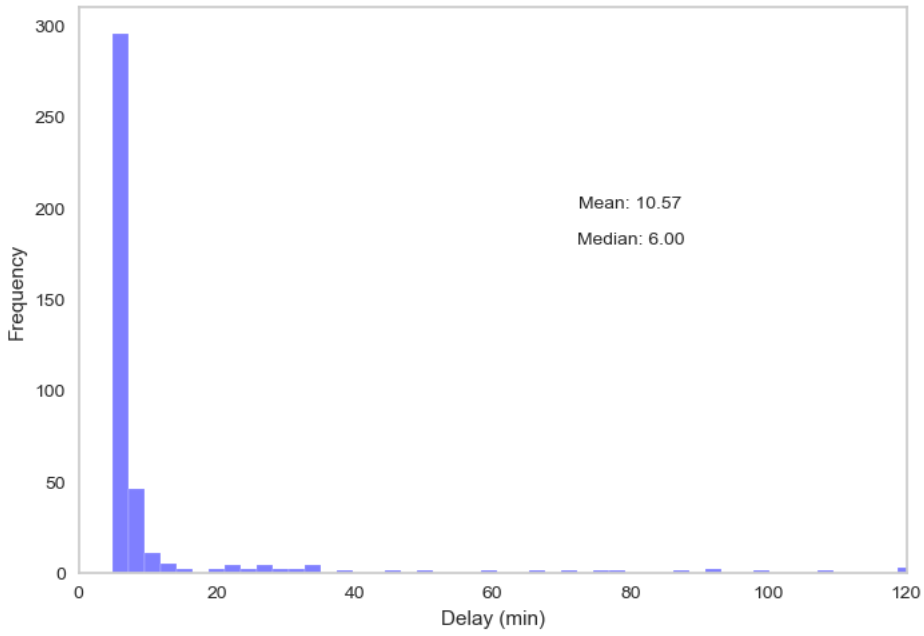


Figure 5.8 Distribution of unlabeled delays due to potential unreported incidents

5.4 Clustering Results

For enhanced clustering results, the selection of cluster numbers is refined using the Silhouette Coefficient (SC). This coefficient measures the similarity of an object to its own cluster relative to other clusters, with a range from -1 to 1. Generally, a higher silhouette score indicates better clustering results, where the clusters are more dense and well-separated from each other. We also explore alternative clustering algorithms such as K-medoids and agglomerative clustering. In K-medoids, clusters are represented by the median of data points instead of the mean, while agglomerative clustering iteratively merges the nearest data points to form clusters.

The performance of K-medoid and agglomerative clustering was systematically compared to K-means, utilizing SC scores as the benchmark for evaluation. After standardizing the input data, clustering was performed, and the results are shown in Figure 5.9. It is evident that K-means had almost the best performance on all different number of clusters. The K-means algorithm achieved its highest SC score when the number of clusters was set to 3, indicating its optimal performance. Thus, a K-means configuration with 3 clusters was chosen as the optimal setup for incident categorization.

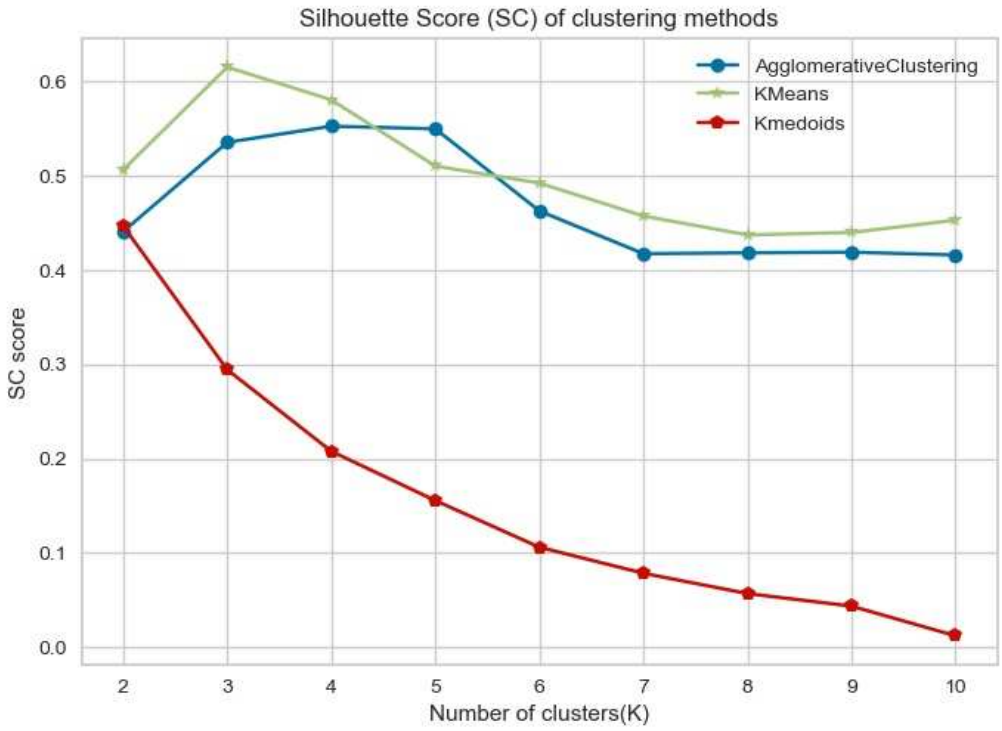


Figure 5.9 Silhouette score of clustering methods

5.4.1 Clusters Characteristics

Each cluster was characterized using statistical measures such as mean and standard deviation (SD) for the input variables. The summarized findings are detailed in Table 5.2, where NAT represents the number of affected trains, NATS signifies the number of affected train-station pairs, TDT denotes total delay time in minutes, and BD equals 1 if both directions are affected. Interpretations of the clusters can be derived from this table as follows:

- *Cluster 1:* This cluster represents the incidents with the least severe impacts. These incidents only affected one direction of a subway line (BD = 0) and resulted in an average total delay time (TDT) of approximately 16 minutes. The average number of affected trains (NAT) was 4.2, with an average delay of nearly 3.8 minutes per train.
- *Cluster 2:* Comprises the majority of incidents. These incidents affected both directions of a subway line (BD=1), resulting in an average TDT of 52.2 minutes. The Average NAT is approximately 12, with an average delay of 4.4 minutes for each train.
- *Cluster 3*: this cluster includes incidents with the most severe impacts on the operation. The incidents within this cluster affected 26 trains and caused a total delay time of 273 minutes on average. Notably, these incidents result in average delays exceeding 10 minutes for each train.

By comparing the statistical measures of different clusters, it can be inferred that cluster 1, cluster 2 and cluster 3 correspond to incidents with low, moderate, and severe impacts respectively.

Table 5.2 Statistical measures of clusters

	Cluster 1 Size = 65		**Cluster 2 Size = 545**		**Cluster 3 Size = 155**	
Impact measure	**Mean**	**SD**	**Mean**	**SD**	**Mean**	**SD**
NAT	4.2	2.8	11.8	4.8	26.1	5.9
NATS	5.0	4.0	14.6	7.1	46.5	20.0
TDT (min)	16.1	12.3	52.2	42.6	273.6	238.3
BD	0.0	0.0	1.0	0.0	1.0	0.0
Avg delay per train (min)	3.8		4.4		10.5	

To gain more insights into the underlying patterns and characteristics of different clusters, density distribution plots of two impact measures including NAT and TDT are presented in Figure 5.10 On the x-axis, the three clusters are represented, while the y-axis illustrates the NAT (left) and TDT (right). The wider portions of the violins indicate a higher concentration of data points within that

range, and the white dot inside each violin represents the median of the y-axis value for each cluster. The findings from Figure 5.10 can be summarized as follows:

- *Number of affected trains (NAT):* The NAT distribution for cluster 1 and cluster 2 appear to be concentrated around 5 and 10 respectively. In contrast, cluster 3, shows a wider spread of NAT, ranging from approximately 20 to 30, with the median around 25.
- *Total delay time (TDT):* Cluster 1 exhibits a lower median and a narrower spread over TDT compared to the other two clusters. This cluster includes minor incidents with low impact in terms of NAT and TDT. For cluster 2, the TDT distribution displays a long tail, with values ranging from approximately 25 to 65, and the median is around 40. In cluster 3, there is a very long-tail distribution for TDT, ranging from 120 to 300. This cluster includes the extreme values for TDT, signifying the presence of infrequent incidents with severe impact (TDT above 300 minutes).

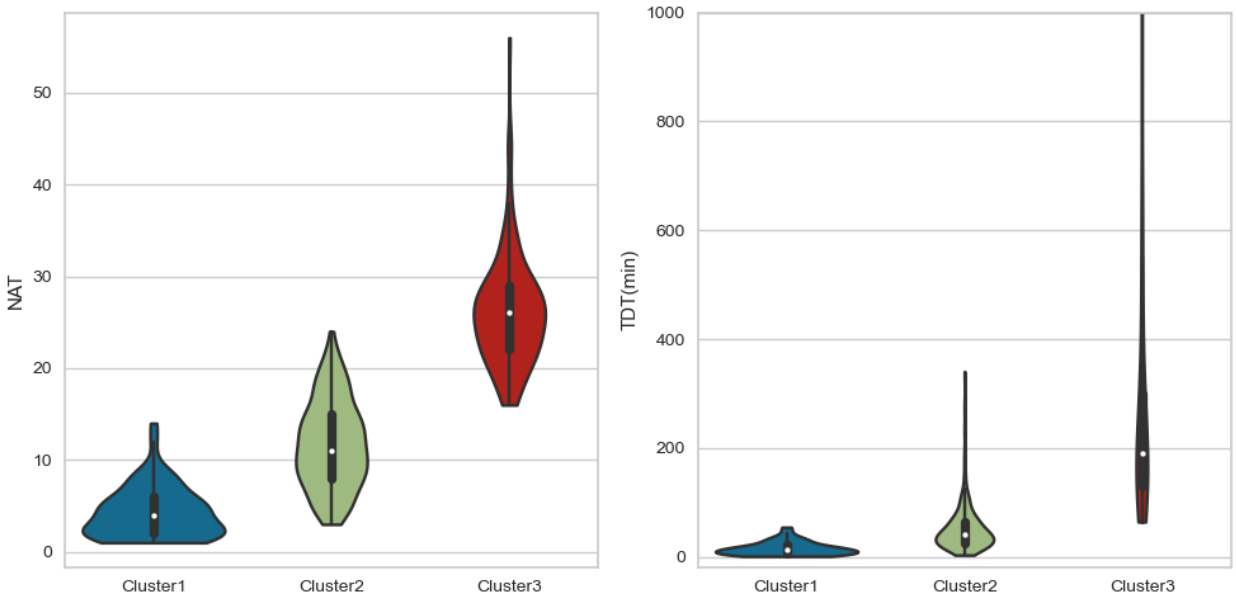


Figure 5.10 Distribution plot of NAT and TDT across different clusters

5.4.2 Temporal Analysis of Clusters

In this section each cluster is examined individually to have a better understanding of the characteristics of incidents within each cluster. Figure 5.11 shows the distribution of primary delay (PD) duration for each cluster across different time periods. The x-axis represents three main service periods: weekday peak hours (from 7 to 9 and 16 to 18), weekday off-peak hours, and

weekends, while the y-axis shows primary delay duration in minutes. Notably, in each time interval, the width of violins is proportional to the number of incidents within that interval.

Cluster 2 has the largest share of incidents across all time periods, suggesting that most incidents had moderate impacts whenever they occurred. Most of the incidents in Cluster 2 had a PD duration of 5 to 20 minutes. Figure 5.11highlights that cluster 1 (low-impact incidents) has a minimal share of incidents occurring during peak hours. This indicates that even small incidents during rush hours mostly belong to cluster 2 and 3 with more severe impacts. Cluster 1 mainly includes incidents with a PD duration of less than 15 minutes, occurring more frequently during off-peak hours (off-peak weekday and weekends). Incidents with a PD duration of more than 20 minutes is predominantly included in cluster 3. The larger the primary delay duration, the less frequent the occurrence, as shown by the narrower width of the violins for Cluster 3.

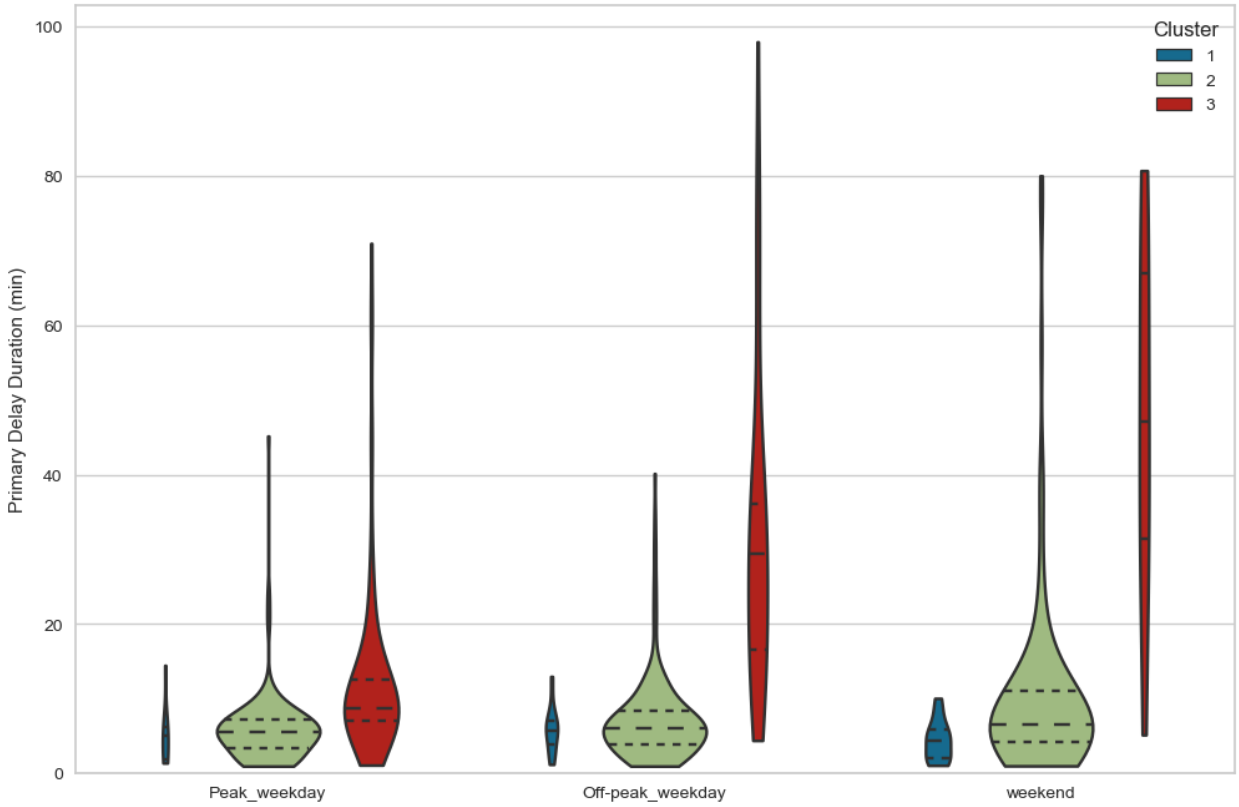


Figure 5.11 Primary delay distributions during peak and off-peak hours

To gain a deeper understanding of the impact of peak hours on the propagation of primary delays, Figure 5.12 presents the distribution of the number of affected trains (NAT) during peak and off-

peak hours. The dashed lines in each violin represent the quantiles, providing a visual representation of the spread of NAT within each cluster.

Notably, higher NAT values are observed during peak hours (from 7 to 9 and 16 to 18), indicating the impact of rush hours on the severity of the incidents. During peak hours service headway is at its lowest level (3 minutes on average), and the density of trains on the tracks is consequently high. As a result, a PD affects more trains compared to off-peak hours when the average headway is more than 6 minutes. For instance, the median of NAT for cluster 3 during off-peak weekday is around 22, while it is 27 during peak hours of weekday. This pattern holds for other clusters as well. Interestingly, for cluster 2, although incidents that occurred on weekends had a larger primary delay duration (according Figure 5.11 compared to weekdays, it had a smaller range for NAT. This is due to lower train frequencies on weekends (every 9 minutes on average) compared to off-peak weekdays (every 6 minutes on average). The same pattern is observed for cluster 3 where the range of NAT for weekends is remarkably less than weekdays. The highest number of affected trains is 27 for weekends, while for weekdays (peak and off-peak), it exceeds 40.

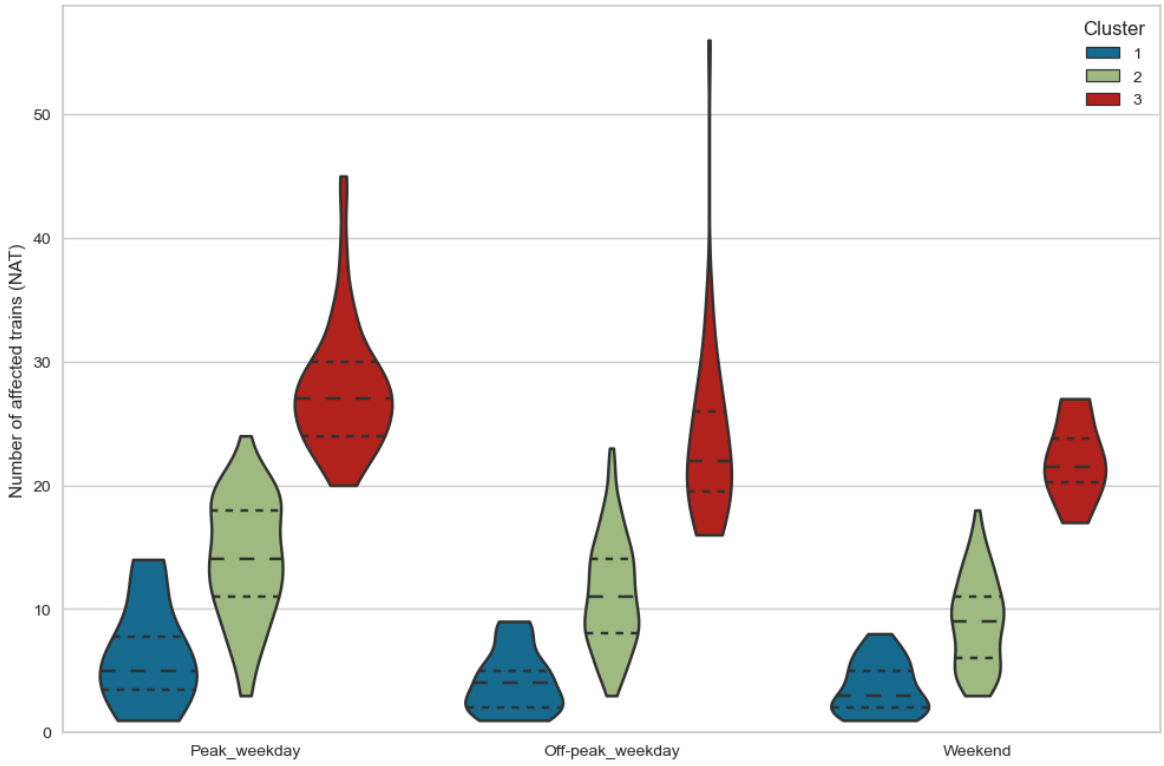


Figure 5.12 Distributions of the number of affected trains during peak and off-peak hours

The distributions for NATS and TDT are shown in Figure 5.13 and Figure 5.14 respectively. Like the distribution of NAT, higher values of NATS are observed during weekdays compared to weekends. The violin plots of cluster 3 show a notably wide range for NATS during weekdays, suggesting that longer incidents result in train delays across multiple stations along their routes. This observation linked to delay recovery strategies where operators stop trains at multiple stations to reduce service gap and keep headways more even.

According to Figure 5.14, the violin plots representing TDT for cluster 3 show narrow distributions with a considerably extensive range compared to the other two clusters. This indicates that the total delay time resulting from severe incidents (cluster 3) demonstrates significant variations and can extend up to several hundred minutes.

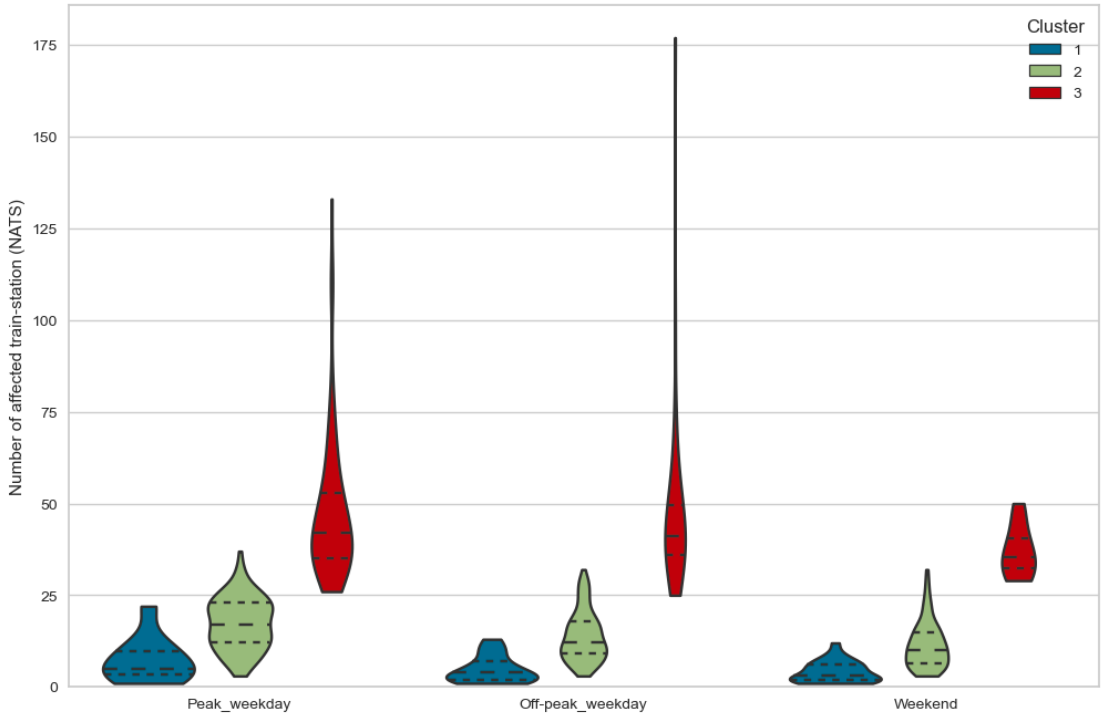


Figure 5.13 Distributions of the number of affected train-stations during peak and off-peak hours

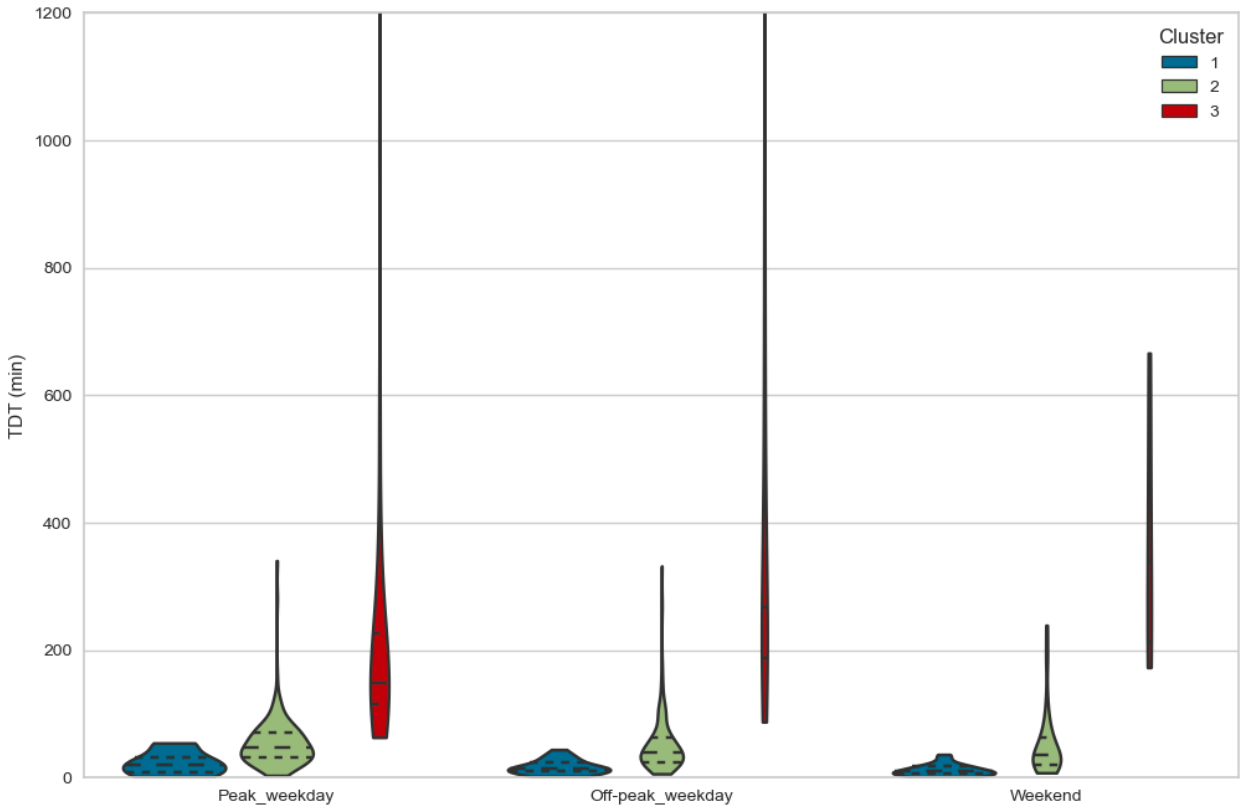


Figure 5.14 Distribution of total delay time (TDT) during peak and off-peak hours

5.4.3 Investigation of Incident Causes

To reveal the influence of different incident causes on subway operation, the most frequent incidents were selected, and their distribution across different clusters is presented in Table 5.3. Each row in the table corresponds to an incident cause, with the percentage indicating the proportion of that incident cause allocated to each cluster. The summary of findings from Table 5.3 is as follows:

- Almost all incident causes have the highest share in cluster 2, indicating that most frequent incidents typically have a moderate impact on subway operation. For instance, 89% of all power failures belong to cluster 2, suggesting that this type of incident generally has a moderate impact.
- Interestingly, the vast majority (83%) of incidents classified as “attempted violent death” belong to cluster 3, indicating severe impacts on subway operations. Only 17% belong to cluster 2, and none belong to cluster 1.

- This table shows that most incidents tend to have a medium impact on subway operations (cluster 2). However, certain incident causes, such as "Attempted violent death," have severe impacts (cluster 3), while others have relatively minor impacts (cluster 1). Understanding these patterns can help in prioritizing resources and improving subway operation management and safety.

Table 5.3 Cluster share of the most frequent incidents

Incident cause	**Number of incidents**	**Cluster 1**	**Cluster 2**	**Cluster 3**
Door failure	150	13%	**64%**	23%
Sick passenger	102	3%	**78%**	19%
Blockage	65	11%	**60%**	29%
Emergency braking	63	5%	**71%**	24%
Power failure	56	4%	**89%**	7%
Object on track	47	4%	**79%**	17%
Equipment failure	41	12%	**66%**	22%
staff related	41	20%	**78%**	2%
Mischief/Nuisance	27	11%	**81%**	7%
Attempted violent death	18	0%	17%	**83%**
police intervention	18	0%	89%	11%
other incidents	137	9%	**72%**	19%
Total	765			

Table 5.4 displays the proportion of the most frequent incidents within each cluster, where total values add up to 100% for each cluster. Top 3 percentages within each cluster are bolded. Door failure incidents have the highest percentages across all clusters, indicating the frequent occurrence of this incident type and the wide range of effects it could have on operation. Incidents caused by sick passengers have the second-highest percentage in clusters 2 and 3, suggesting this type of incident can lead to severe impacts on operation. Interestingly, staff-related incidents have the second-highest share within cluster 1, while it has the lowest share in cluster 3.

Table 5.4 Distribution of the most frequent incidents within each cluster

Incident cause	**Cluster 1**	**Cluster 2**	**Cluster 3**
Door failure	**29%**	**18%**	**23%**
Sick passenger	5%	**15%**	**12%**
Blockage	**11%**	7%	**12%**
Emergency brake	5%	8%	10%
Power failure	3%	**9%**	3%
Staff related	**12%**	6%	1%
Object on track	3%	7%	5%
Equipment failure	8%	5%	6%
Mischief/Nuisance	5%	4%	1%
Attempted violent death	0%	1%	10%
police intervention	0%	3%	1%
Other incidents	20%	18%	17%
Total	100%	100%	100%

CHAPTER 6 CONCLUSION AND RECOMMENDATIONS

6.1 Summary of the Main Findings

The work presented in this thesis was dedicated to developing a method for analyzing incident effects on subway operation. The proposed approach leveraged the operational data from the Montréal subway system to identify incident severity patterns through a clustering analysis. A review of the existing literature on subway incident management revealed a few gaps that determined the main objectives of this research. The first objective was to develop an algorithm to process raw operational data and gives the detailed trajectories of subway train services. The second objective focused on developing an approach for characterizing incidents using incident severity indicators. As an important output, this research delivers a comprehensive analysis of the propagation of incident delays along subway lines which enables us to distinguish primary and secondary delays and to measure the full effects of incidents on operation.

This chapter gives a summary of the main findings and practical contributions of this research. Additionally, limitations and recommendations for future research are given in section 6.2.

6.1.1 Performance Evaluation of Subway Service

For the Montréal subway system, train positioning data is sourced from fixed block signalling system of variable length that covers all tracks within the network. This technology, combined with an autopilot system, enables precise recording of the exact time a train occupies and vacates a block, down to the second. Thus, the first objective of this research was to develop a data mining algorithm that can quickly process large blocks data and extract train paths (trajectories). The outcome of the work is a data mining tool capable of reconstructing daily trajectories of a subway line in just a few seconds. The resulting data structure is convenient for statistical analysis and model calibration in this and other research projects. It enables various analyses, including service headway and punctuality assessment, dwell time and travel time analysis, and the creation of speed profiles, among others.

6.1.2 Measuring Incident Effects

The second research objective in this thesis was directed at measuring incident effects on subway operation. To do so, station occupancy time (SOT) was considered as an indicator for quantifying train delays at stations. It was shown that the standard operational procedure for managing incident

delays in the Montréal subway network is to stop trains at the stations. Thus, the deviation of SOT from a normal SOT would give train delays. Subsequently, incidents and their resulting knock-on delays were identified in trajectory data using incident logs. Incident impact (severity) indicators were defined to help in aggregating the full effects of each incident. Finally, severity indicators were fed into K-means clustering to categorize incidents based on their impact.

The Clustering analysis revealed three distinct groups of incidents with varying degrees of impact within the Montréal subway system. Cluster 1 comprised incidents with minimal impacts, only affecting one direction of a subway line, imposing an average total delay time of 16 minutes. Cluster 2, representing most incidents, having medium impacts on both directions of subway lines, resulting in an average total delay time of 52 minutes, and affecting approximately 12 trains per incident. Cluster 3 included incidents with severe consequences, significantly disrupting subway services by affecting 26 trains on average and causing a total delay time of 273 minutes. The influence of peak hours on incident propagation was evident, with morning and evening peak hours showing higher average numbers of affected trains. Rush hours, characterized by reduced service headway, led to incidents impacting a larger number of trains compared to off-peak hours and weekends. Furthermore, an investigation into the incident causes revealed that most frequent incidents tend to have moderate impacts on subway operation.

The comparison between delays reported in incident logs and those identified in trajectories data revealed a potential bias towards overestimation of primary delays in incident logs, especially for delays less than 20 minutes. This highlights the necessity of validating data from multiple sources to enhance the precision of predictive models, particularly concerning smaller delays. Moreover, the presence of unlabelled delays exceeding 5 minutes in trajectories data indicates potential incidents overlooked in incident logs, emphasizing the inadequacy of relying solely on incident logs for a comprehensive analysis of incident delays.

Understanding incident severity patterns enables transit operators to enhance their knowledge about the full effects of incidents and improve decision-making for efficient resource allocation and subway operation management. Our findings allow operators to assess the effectiveness of their mitigation strategies in incident delay management. For instance, the freezing strategy commonly employed in the Montréal subway system may not be the best recovery measure for all incident

types. By evaluating the impact of various mitigation strategies in different operational contexts, operators can adjust their response strategies to minimize the consequences.

6.2 Limitations and Recommendations Future Work

This study contributes valuable insights into subway incident management, providing a basis for future research and the development of effective strategies to mitigate disruptions and improve the overall performance of urban rail transit systems. However, certain limitations should be acknowledged. We recognize the challenge of establishing causal relationships between incident causes and their consequences in our study. Our analysis primarily aims to explore associations and patterns within the data, rather than inferring a causal relationship.

For measuring incident impacts, we only relied on those incidents reported by the operator in incident logs. Our analysis showed the existence of potentially unreported incidents which highlights the limitation of such data. However, the cause of those unreported delays is unknown and needs further investigation. Future research should focus on merging more data sources such as smart card data, social media, etc. to infer the cause of delays.

This study lacks a passenger-oriented perspective due to the absence of information on the number of subway passengers during the study period. It is recommended that future research incorporate passenger-oriented indicators, such as the number of affected passengers and passenger hours of delay, to assess the impact of incidents on subway users. Furthermore, integrating spatial analysis into future studies could provide valuable insights into the relationship between station characteristics and incident severity, further enriching our understanding of the dynamics within subway systems.

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